

# Exact exponential periods, invariant constituent curvature, and collision homology

Split-Zero research programme

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## 1 Exact determinant of the polynomial–exponential period comparison

Fix the original monic cyclic polynomial

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q \geq 1, \quad d = q + 1, \quad \Phi_t(S) = \frac{S^d}{d} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS.$$

Fix  $u \in \mathbb{C}^*$  and one argument of  $u$ . All the following comparisons retain this  $u$ , the original coordinate  $S$ , and the ordered remainder basis  $1, S, \dots, S^{q-1}$ . Set

$$D_{u,t}P = uP' + (\chi - t)P, \quad H_{u,t} = \mathbb{C}[S]/D_{u,t}\mathbb{C}[S].$$

This is a quotient of coefficient vector spaces. Let  $\text{red}_{u,t}$  be its unique representative of degree less than  $q$ . Let  $A(t)$  be the multiplication matrix of  $S$  on  $\mathbb{C}[S]/(\chi - t)$  in the same ordered basis, so

$$A(t)e_b = e_{b+1} \quad (0 \leq b < q-1), \quad A(t)e_{q-1} = (-c_0 + t)e_0 - \sum_{a=1}^{q-1} c_a e_a. \quad (\text{XD1})$$

The empty sum convention in this display retains the case  $q = 1$ .

Use precisely the oriented rays and differences

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \ell_j : S = re^{i\theta_j}, \quad 0 \leq r < \infty, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq q). \quad (\text{XD2})$$

Write

$$\Pi_{jb}(u, c, t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{XD3})$$

### 1.1 Uniform estimates on the fixed unbounded contours

For a compact set  $K \subset \mathbb{C}^{q+1}$  of the parameters  $(c_0, \dots, c_{q-1}, t)$ , put

$$M_K = \sum_{a=0}^{q-1} \frac{\sup_K |c_a|}{a+1} + \sup_K |t|, \quad R_K = \max\{1, 2dM_K\}, \quad \alpha = \frac{1}{2d|u|}. \quad (\text{XD4})$$

For every ray in (XD2) its leading term satisfies  $S^d/(du) = -r^d/(d|u|)$ . For  $r \geq 1$  every other exponent has degree at most  $d-1$ , so

$$\text{Re} \frac{\Phi_t(re^{i\theta_j})}{u} \leq -\frac{r^d}{d|u|} + \frac{M_K r^{d-1}}{|u|} \leq -\alpha r^d \quad (r \geq R_K). \quad (\text{XD5})$$

Every parameter derivative of the integrand is its original exponential times a polynomial in  $S$ : a derivative in  $c_a$  contributes  $S^{a+1}/((a+1)u)$  and a derivative in  $t$  contributes  $-S/u$ . For any fixed derivative order its absolute value on the tail is bounded by  $Cr^Le^{-\alpha r^d}$ , with  $C, L$  uniform on  $K$ . The required majorant is integrable, since substitution  $x = \alpha r^d$  gives

$$\int_0^\infty r^L e^{-\alpha r^d} dr = \frac{1}{d} \alpha^{-(L+1)/d} \Gamma\left(\frac{L+1}{d}\right) < \infty. \quad (\text{XD6})$$

The remaining finite interval is uniformly bounded by continuity. Thus (XD3) converges absolutely and is entire in all the lower coefficients and  $t$ ; every parameter derivative may be taken under the integral. The same bound gives  $e^{\Phi_t(S)/u} P(S) \rightarrow 0$  at the infinite endpoint of each ray, for every polynomial  $P$ . Consequently

$$\int_{\Gamma_j} e^{\Phi_t/u} D_{u,t} P dS = u \int_{\Gamma_j} d(e^{\Phi_t/u} P) = 0. \quad (\text{XD7})$$

Indeed the integral of that derivative on each outward ray is  $-e^{\Phi_t(0)/u} P(0) = -P(0)$ ; the coefficients  $-1, +1$  of the two rays cancel these values. These estimates handle the noncompact contours directly, including parameter values at which critical points coincide.

## 1.2 An exact trace comparison for twisted reduction

For  $1 \leq m \leq q$  and  $0 \leq b < q$ , ordinary monic division gives unique polynomials  $Q_{m,b}, R_{m,b}$  satisfying

$$S^{m+b} = (\chi - t)Q_{m,b} + R_{m,b}, \quad \deg R_{m,b} < q. \quad (\text{XD8})$$

If  $m+b < q$ , take  $Q_{m,b} = 0$ . Otherwise  $\deg Q_{m,b} = m+b-q \leq q-1$ , and the differential itself gives

$$\text{red}_{u,t}(S^{m+b}) = R_{m,b} - uQ'_{m,b}. \quad (\text{XD9})$$

Both terms on the right have degree below  $q$ . In every nonzero case,

$$\deg Q'_{m,b} \leq m+b-q-1 \leq b-1. \quad (\text{XD10})$$

Thus the coefficient of  $S^b$  in the derivative correction vanishes. Let  $T_m(u, c, t)$  have column  $b$  equal to the coordinates of the left side of (XD9). The columns  $R_{m,b}$  are exactly those of  $A(t)^m$ . Although (XD9) retains a possibly nonzero correction matrix, (XD10) proves the exact trace identity

$$\text{Tr } T_m(u, c, t) = \text{Tr } A(t)^m \quad (1 \leq m \leq q). \quad (\text{XD11})$$

Set  $B_a = T_{a+1}/((a+1)u)$  for  $0 \leq a < q$ . Differentiation of (XD3), followed by (XD7)–(XD9), now proves

$$\partial_{c_a} \Pi = \Pi B_a, \quad \partial_t \Pi = -\Pi A(t)/u, \quad \text{Tr } B_a = \frac{\text{Tr } A(t)^{a+1}}{(a+1)u}. \quad (\text{XD12})$$

All the coefficient matrices here have polynomial entries in  $c, t$  for this fixed nonzero  $u$ .

## 1.3 The determinant, including every monomial factor

Define the scalar polynomial in the original coefficients

$$\mathcal{F}(c, t) = \text{Tr } \Phi_t(A(t)) = \frac{\text{Tr } A(t)^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a \text{Tr } A(t)^{a+1}}{a+1} - t \text{Tr } A(t). \quad (\text{XD13})$$

There is no choice or enumeration of roots in this definition. For any differentiable matrix  $A$  and integer  $n \geq 1$ , the product rule and cyclic invariance of trace give

$$d \operatorname{Tr} A^n = \sum_{r=0}^{n-1} \operatorname{Tr}(A^r (dA) A^{n-1-r}) = n \operatorname{Tr}(A^{n-1} dA). \quad (\text{XD14})$$

Because  $A(t)$  acts on the actual quotient by  $\chi - t$ , it satisfies  $\chi(A(t)) - tI_q = 0$ . Applying (XD14) to (XD13) therefore gives

$$d\mathcal{F} = \sum_{a=0}^{q-1} \frac{\operatorname{Tr} A(t)^{a+1}}{a+1} dc_a - \operatorname{Tr} A(t) dt + \operatorname{Tr}((\chi(A(t)) - tI_q) dA(t)) = \sum_{a=0}^{q-1} \frac{\operatorname{Tr} A(t)^{a+1}}{a+1} dc_a - \operatorname{Tr} A(t) dt. \quad (\text{XD15})$$

At  $c = t = 0$  the period matrix is

$$\Pi_{j,l-1}^{\text{mon}}(u) = a_l(u) e^{il\theta_0} (\zeta^{jl} - 1), \quad a_l(u) = \frac{(d|u|)^{l/d}}{d} \Gamma(l/d), \quad \zeta = e^{2\pi i/d}, \quad 1 \leq j, l \leq q. \quad (\text{XD16})$$

This follows by substituting  $x = r^d/(d|u|)$  on each oriented ray in (XD3). In particular all  $a_l(u)$  are positive real numbers. For  $V_{jl} = \zeta^{jl} - 1$  one may include  $j = 0$  in the Gram sum because its summand is zero. The complete root-of-unity sums give

$$(V^*V)_{lm} = \sum_{j=0}^{d-1} (\zeta^{-jl} - 1)(\zeta^{jm} - 1) = d\delta_{lm} + d. \quad (\text{XD17})$$

Here  $1 \leq l, m < d$ , so the two individual nonconstant sums vanish, and the first sum is  $d\delta_{lm}$ . The all-ones rank-one matrix has eigenvalue  $q$  on the all-ones vector and zero on its  $(q-1)$ -dimensional perpendicular complement. It follows that

$$\det(V^*V) = d^q(1+q) = d^d > 0, \quad C_q(u) := |\det \Pi^{\text{mon}}(u)|^2 = d^d \prod_{l=1}^q \left[ \frac{(d|u|)^{l/d}}{d} \Gamma(l/d) \right]^2. \quad (\text{XD18})$$

The complex determinant itself retains the oriented phase:  $\det \Pi^{\text{mon}} = \det V \prod_{l=1}^q a_l(u) e^{il\theta_0}$ .

We claim the exact global formula

$$\boxed{\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp(\mathcal{F}(c, t)/u)}. \quad (\text{XD19})$$

To prove it without assuming invertibility along the path, join  $(0, 0)$  to  $(c, t)$  by  $(sc, st)$ ,  $0 \leq s \leq 1$ . Equations (XD12) give  $\Pi' = \Pi B(s)$ , with continuous  $B(s)$ , and (XD15) gives  $\operatorname{Tr} B(s) = \mathcal{F}'(sc, st)/u$ . For every square matrix, including a singular one,

$$\frac{d}{ds} \det \Pi = \operatorname{Tr}(\operatorname{adj}(\Pi) \Pi B) = \det \Pi \operatorname{Tr} B. \quad (\text{XD20})$$

Consequently  $\det \Pi \exp(-\mathcal{F}/u)$  has derivative zero. At the starting point  $A$  is nilpotent and  $\mathcal{F}(0, 0) = 0$ , so its value is  $\det \Pi^{\text{mon}}(u)$ . This proves (XD19). Equation (XD18) proves its nonvanishing for every  $c, t$ , including every repeated-root fibre. In particular the period map

$$H_{u,t} \longrightarrow \mathbb{C}^q, \quad [P] \longmapsto \Pi \operatorname{red}_{u,t}(P), \quad (\text{XD21})$$

is an isomorphism with inverse given by  $\Pi^{-1}$  followed by the specified degree-below- $q$  section.

## 1.4 Exact determinant transport to the arithmetic representative

Retain the original positive Gram  $G_N$  on the cyclic coefficient space  $E$ , its determinant  $V_N = \det G_N$ , and the original maps  $r_N : E \rightarrow \mathcal{B}^{\otimes k}$  and  $j_E$  on the admitted test-function domain with  $j_E r_N = I_E$ . Identify  $E$  with  $H_{u,t}$  by their specified ordered remainder bases. Then the actual maps

$$r_N(E) \rightarrow \mathbb{C}^q, \quad x \mapsto \Pi j_E x, \quad \mathbb{C}^q \rightarrow r_N(E), \quad y \mapsto r_N \Pi^{-1} y \quad (\text{XD22})$$

are inverse: their composites are respectively  $r_N j_E$  on  $r_N(E)$  and  $\Pi(j_E r_N) \Pi^{-1}$  on  $\mathbb{C}^q$ . On the full admitted domain the first observation has precisely  $\ker j_E$  as its kernel; all original theta boundaries killed by  $j_E$  remain in that kernel.

With the explicitly fixed coordinate metric  $I_q$  on the target, put

$$H_\Pi = \Pi^* \Pi, \quad B_N^{\text{per}} = G_N^{-1} H_\Pi. \quad (\text{XD23})$$

These operators satisfy  $G_N B_N^{\text{per}} = (B_N^{\text{per}})^* G_N = H_\Pi > 0$ . Taking determinants in (XD19) gives the exact identities

$$\det H_\Pi = C_q(u) \exp\left(2 \operatorname{Re} \frac{\mathcal{F}(c, t)}{u}\right), \quad (\text{XD24})$$

$$\boxed{\log \det B_N^{\text{per}} = \log C_q(u) + 2 \operatorname{Re} \frac{\mathcal{F}(c, t)}{u} - \log V_N.} \quad (\text{XD25})$$

Every determinant in (XD24)–(XD25) is positive real; these logarithms therefore require no branch choice. At the same  $u, c, t$  for two admitted degrees  $i \leq j$ , subtraction proves

$$\boxed{\log \det B_j^{\text{per}} - \log \det B_i^{\text{per}} = \log(V_i/V_j).} \quad (\text{XD26})$$

Thus the common polynomial–exponential factor is now calculated exactly while the full original endpoint cost is retained by the displayed pair of positive comparison operators. No uniform upper estimate on  $V_i/V_j$  is inferred from this determinant identity.

## 2 SGA trace, exact constituent curvature, and the derived collision fibre

This calculation extends the constituent period map in the delivered *SGA Trace Period Control*, equations (54a)–(54c). Its period determinant is the same determinant already proved in (XD1)–(XD26), with every original coefficient, ray and phase retained. We calculate the full normal derivative, its kernel and its norm. The result applies to the actual invariant constituents, including repeated primary factors.

### 2.1 The retained period map and arithmetic source

Keep the original monic polynomial and its increasing-power frame:

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S, \quad q \geq 1, \quad e_0 = [1], \quad \ell([P]) = [S^{q-1}] \operatorname{rem}_\chi P. \quad (\text{SC1})$$

Thus  $\ell : E \rightarrow \mathbb{C}$  is a specified linear functional and  $\mathcal{R} = e_0 \ell : E \rightarrow E$  has rank one. Fix  $u \in \mathbb{C}^*$ , one argument of  $u$ , and the actual potential and contours

$$\Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS, \quad \theta_j = \frac{\arg u + \pi + 2\pi j}{q+1}, \quad \Gamma_j = -\ell_0^{\text{ray}} + \ell_j^{\text{ray}}. \quad (\text{SC2})$$

Here  $\ell_j^{\text{ray}}$  is the outward ray  $S = re^{i\theta_j}$ ,  $r \geq 0$ ; it is distinct from the coefficient functional  $\ell$ . The period matrix is

$$\Pi_{jb}(t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{SC3})$$

The full convergence and determinant proof is (XD4)–(XD21). To state its precise retained inputs here, on a compact coefficient set  $K$  its tail is bounded by  $r^L \exp(-r^{q+1}/(2(q+1)|u|))$  after  $r \geq \max(1, 2(q+1)M_K)$ , where  $M_K = \sum_a \sup_K |c_a|/(a+1) + \sup_K |t|$ . Differentiation is therefore valid in every coefficient direction. Integration by parts cancels the two origin terms, and reduction of  $S^q$  uses  $[\chi - t] = 0$  with no derivative correction. It follows that

$$\Pi'(t) = -\frac{1}{u} \Pi(t)(A + t\mathcal{R}). \quad (\text{SC4})$$

The proven determinant is

$$\det \Pi(t) = \det \Pi_{\text{mon}}(u) \exp\left(\frac{\text{Tr } \Phi_t(A + t\mathcal{R})}{u}\right), \quad |\det \Pi_{\text{mon}}(u)|^2 = (2\pi|u|)^q > 0. \quad (\text{SC5})$$

The complex initial matrix is explicitly

$$(\Pi_{\text{mon}})_{j,b} = \frac{((q+1)|u|)^{(b+1)/(q+1)}}{q+1} \Gamma\left(\frac{b+1}{q+1}\right) e^{i(b+1)\theta_0} (e^{2\pi i j(b+1)/(q+1)} - 1). \quad (\text{SC6})$$

The product form and the full complex phase in (SC5)–(SC6) are reused from (XD18)–(XD19). The evaluated squared constant is the additional gamma-product evaluation in the delivered SGA note. Here is every factor in that evaluation. Set  $d = q + 1$ . The matrix  $V_{jl} = e^{2\pi i j l/d} - 1$  has  $V^*V = d(I_q + J_q)$  by the root-of-unity sum in (XD17), so  $|\det V|^2 = d^{q+1}$ . Since  $2 \sum_{l=1}^{d-1} l/d = q$ , its column factors give

$$|\det \Pi_{\text{mon}}|^2 = d^{q+1} \frac{(d|u|)^q}{d^{2q}} \left( \prod_{l=1}^{d-1} \Gamma(l/d) \right)^2. \quad (\text{SC6a})$$

The gamma multiplication formula, DLMF (5.5.6), is

$$\Gamma(dz) = (2\pi)^{(1-d)/2} d^{dz-1/2} \prod_{a=0}^{d-1} \Gamma(z + a/d). \quad (\text{SC6b})$$

At  $z = 1/d$ , its left side and the final product factor are  $\Gamma(1) = 1$ , so the remaining product equals  $(2\pi)^{q/2} d^{-1/2}$ , which is DLMF (5.5.7). In (SC6a) the complete power of  $d$  is  $(q+1) + q - 2q - 1 = 0$ . This proves  $|\det \Pi_{\text{mon}}|^2 = (2\pi|u|)^q$  without a frame change or an omitted phase. Thus every  $\Pi(t)$  is invertible, including at collisions. The full reference is <https://dlmf.nist.gov/5.5.E7>.

For every original admitted source degree  $N$ , retain

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad r_N = \mathcal{V} R_N, \quad j_E r_N = I_E. \quad (\text{SC7})$$

The source identities and primitive have their original types  $j_E D^{(k)} = A j_E$  and  $D^{(k)} r_N - r_N A = d K_N^{\text{prim}}$ . The exact parameter action on the admitted source domain is

$$D_t^{(k)} = D^{(k)} + t r_N \mathcal{R} j_E, \quad j_E D_t^{(k)} = (A + t\mathcal{R}) j_E, \quad D_t^{(k)} r_N - r_N (A + t\mathcal{R}) = d K_N^{\text{prim}}. \quad (\text{SC8})$$

Indeed insertion of  $j_E r_N = I_E$  proves each equality by multiplication; the added terms cancel in the last difference. In particular the normal map calculated below has an explicit lift to this source, with the complete original primitive and kernel of  $j_E$  retained.

## 2.2 Every invariant constituent has a nonzero top coefficient

Let  $I : F \hookrightarrow E$  be an actual injective linear map whose image is  $A$ -invariant, with  $AI = IA_F$ , and put  $p = \dim F$ . For the moment suppose  $0 < p < q$ . The image  $I(F)$  is an ideal of  $E$ : invariance under  $A$  implies invariance under every polynomial in  $A$ , and every element of  $E$  is such a polynomial applied to  $e_0$ . Its inverse image in  $\mathbb{C}[S]$  is an ideal containing  $(\chi)$ . It is generated by a unique monic divisor  $g$  of  $\chi$ ; writing  $\chi = gh$  gives  $\deg h = p$  and  $\deg g = q - p$ . The exact map is

$$\mathbb{C}[S]/(h) \longrightarrow I(F), \quad [a]_h \longmapsto [ga]_\chi. \quad (\text{SC9})$$

It is onto by the definition of  $(g)$ , and is injective because  $gh \mid ga$  in the polynomial domain implies  $h \mid a$ . Consequently  $[g], [Sg], \dots, [S^{p-1}g]$  is an ordered basis of  $I(F)$ . Its last member is monic of degree  $q - 1$ , while every earlier member has smaller degree. Hence the restricted functional  $L = \ell I : F \rightarrow \mathbb{C}$  is onto. Also  $e_0 \notin I(F)$ , since an ideal containing the unit is the whole algebra. We have proved

$$\begin{aligned} \mathcal{R}I &= e_0 L, \quad \text{rank}(\mathcal{R}I) = 1, \quad \ker(\mathcal{R}I) = \ker L, \quad \dim \ker L = p - 1, \\ \mathbb{C}e_0 \cap I(F) &= \{0\} \quad \text{inside } E. \end{aligned} \quad (\text{SC10})$$

These arguments never replace a repeated factor by its square-free part. In the basis (SC9),  $L$  is exactly the last coordinate; in the original chosen basis of  $F$  it remains the literal row  $\ell I$ .

## 2.3 The exact normal derivative at every parameter

Give the target  $\mathbb{C}^q$  the period-coordinate Hermitian metric  $I_q$  specified in the delivered note. Define

$$\begin{aligned} Y(t) &= \Pi(t)I, \quad H(t) = Y(t)^* Y(t), \quad P(t) = Y(t)H(t)^{-1}Y(t)^*, \\ N(t) &= I_q - P(t), \quad z(t) = N(t)\Pi(t)e_0. \end{aligned} \quad (\text{SC11})$$

Every  $Y(t)$  is injective, so  $H(t) > 0$ . Multiplication shows  $P(t)^2 = P(t) = P(t)^*$  and  $P(t)Y(t) = Y(t)$ ; thus  $N(t)$  is the orthogonal projection onto the complement of the displayed period subspace. The vector  $z(t)$  is nonzero for every  $t$ : otherwise  $\Pi(t)e_0$  lies in  $\Pi(t)I(F)$ , and invertibility of  $\Pi(t)$  would give  $e_0 \in I(F)$ , contradicting (SC10).

Using invariance under the original  $A$ , equation (SC4) yields

$$Y'(t) = -\frac{1}{u}Y(t)A_F - \frac{t}{u}\Pi(t)e_0L, \quad \boxed{N(t)Y'(t) = -\frac{t}{u}z(t)L.} \quad (\text{SC12})$$

This holds for every complex  $t$ , with the same fixed constituent  $F$  of the original  $A$ . We have not assumed that  $F$  is invariant under  $A + t\mathcal{R}$ . Formula (SC12) computes exactly its failure of invariance after passing through the original periods. For  $t \neq 0$  the normal derivative has image  $\mathbb{C}z(t)$ , kernel  $\ker L$  and rank one. At  $t = 0$  it vanishes. Differentiating (SC4) once more gives

$$Y''(0) = \frac{1}{u^2}\Pi(0)A^2I - \frac{1}{u}\Pi(0)\mathcal{R}I, \quad N(0)Y''(0) = -\frac{1}{u}z(0)L. \quad (\text{SC13})$$

The first term has image inside  $Y(0)F$  because  $A^2I = IA_F^2$ . The normal acceleration therefore has rank exactly one, image  $\mathbb{C}z(0)$  and kernel  $\ker L$ , including all repeated-root cases.

The actual original source lift of its parameter defect is  $r_N e_0 L : F \rightarrow r_N(E)$ . The map  $x \mapsto N(t)\Pi(t)j_E x$  sends it to  $z(t)L$ . Thus its source, coefficient and period forms have the commuting factorization

$$F \xrightarrow{L} \mathbb{C} \xrightarrow{a \mapsto ar_N e_0} r_N(E) \xrightarrow{j_E} E \xrightarrow{N(t)\Pi(t)} \mathbb{C}^q. \quad (\text{SC14})$$

At the original Split-Zero support, every linear arrow lifts by  $x^\bullet \mapsto f(x)^\bullet$  and  $\tau \mapsto \tau$ . In particular  $\ker L$  maps to the supported zero; external absence retains its distinct image  $\tau$ . No new zero-adjunction is used.

## 2.4 Curvature is an exact rank-one expression

Write  $\partial_t = \frac{1}{2}(\partial_x - i\partial_y)$  and  $\partial_{\bar{t}} = \frac{1}{2}(\partial_x + i\partial_y)$  for  $t = x + iy$ . Since  $Y$  is holomorphic,  $\partial_t H = Y^* Y'$ ,  $\partial_{\bar{t}} H = Y'^* Y$  and  $\partial_{\bar{t}} \partial_t H = Y'^* Y'$ . Differentiating  $H^{-1} H = I_p$  gives  $\partial_{\bar{t}} H^{-1} = -H^{-1} Y'^* Y H^{-1}$ . The determinant derivative and cyclic invariance of trace therefore give

$$\begin{aligned} \partial_{\bar{t}} \partial_t \log \det H &= \text{Tr}(H^{-1} Y'^* Y' - H^{-1} Y'^* Y H^{-1} Y^* Y') \\ &= \text{Tr}(H^{-1} Y'^* N Y'). \end{aligned} \quad (\text{SC15})$$

There is no logarithm-branch choice:  $\det H > 0$ . Insert the full normal map (SC12), use  $N^* = N = N^2$ , and retain the rank-one factors to obtain

$$\partial_{\bar{t}} \partial_t \log \det H(t) = |t|^2 c_F(t), \quad c_F(t) = \frac{\|z(t)\|_{L_q}^2}{|u|^2} L H(t)^{-1} L^* > 0. \quad (\text{SC16})$$

Strict positivity follows because  $z(t) \neq 0$ ,  $L \neq 0$  and  $H(t)^{-1} > 0$ . Thus this curvature vanishes exactly at  $t = 0$  and vanishes there to precisely quadratic order; it is strictly positive at every  $t \neq 0$ . For the dual determinant line with metric  $(\det H)^{-1}$  the convention  $c_1 = -(i/(2\pi)) \partial \bar{\partial} \log h$  gives

$$c_1 = \frac{i}{2\pi} |t|^2 c_F(t) dt \wedge d\bar{t}. \quad (\text{SC17})$$

In these coordinates  $i dt \wedge d\bar{t} = 2 dx \wedge dy$ ; this records the orientation and factor explicitly.

For completeness the exact formula supplies a convergent Taylor statement. Entries of  $\Pi$  have convergent complex power series; entries of  $H$ ,  $P$ ,  $N$  and  $c_F$  are real analytic because  $\det H$  is everywhere positive. On any closed disk  $|t| \leq \varepsilon$ , continuity of the first real derivatives gives a finite bound  $M_\varepsilon = \sup \|\nabla c_F\|$ . Integrating the derivative along the real segment  $s \mapsto st$  proves  $|c_F(t) - c_F(0)| \leq M_\varepsilon |t|$ . Consequently

$$\left| \partial_{\bar{t}} \partial_t \log \det H(t) - |t|^2 c_F(0) \right| \leq M_\varepsilon |t|^3. \quad (\text{SC18})$$

This estimate follows from the actual analytic functions in (SC11), and (SC16) supplies the stronger identity at all parameters.

One may check the projector's first derivatives directly:  $\partial_t P = N Y' H^{-1} Y^*$  and  $\partial_{\bar{t}} P = Y H^{-1} Y'^* N$ . They vanish at zero by (SC12), explaining the retained stationary tangent space at that parameter without deleting its acceleration.

## 2.5 Both original theta and period metrics remain explicit

On  $F$  retain the positive original theta Gram  $G_{N,F} = I^* G_N I$  and the positive comparison operator

$$B_{N,F}(t) = G_{N,F}^{-1} H(t), \quad \mathcal{O}_{N,F}(t) = \log \frac{\det H(t)}{\det G_{N,F}}. \quad (\text{SC19})$$

Here  $G_{N,F}$  is fixed as  $t$  varies. Thus (SC15)–(SC18) hold unchanged with  $\mathcal{O}_{N,F}$  in place of  $\log \det H$ . For real  $t$  at zero and  $u > 0$ , (SC12) gives  $H'(0) = -(A_F^* H(0) + H(0) A_F)/u$ , hence

$$-u \mathcal{O}'_{N,F}(0) - kp = 2 \text{Re Tr } A_F - kp. \quad (\text{SC20})$$

The original constant  $k$  and its generalized eigenspace multiplicities are retained. On the full positive-real-defect constituent this is the existing aggregate defect  $L_{h,k}$ .

The squared Hilbert–Schmidt norm of the normal acceleration in the original domain metric, and its one nonzero squared singular value, are both

$$a_{N,F} = \frac{\|z(0)\|_{I_q}^2}{|u|^2} L G_{N,F}^{-1} L^*. \quad (\text{SC21})$$

Indeed the adjoint of  $zL : (F, G_{N,F}) \rightarrow (\mathbb{C}^q, I_q)$  is  $G_{N,F}^{-1} L^* z^*$ , so the product and its trace give (SC21). The unique nonzero eigenvalue equals its trace because the product has rank one. The curvature coefficient is instead the exact observer

$$c_F(t) = \frac{\|z(t)\|_{I_q}^2}{|u|^2} L B_{N,F}(t)^{-1} G_{N,F}^{-1} L^*. \quad (\text{SC22})$$

This follows from  $H = G_{N,F} B_{N,F}$ ; it is the required map between the two metric measurements. It does not replace either metric by an unweighted coordinate norm.

Finally transport the original metric on the whole period target by  $\tilde{G}_N = \Pi^{-*} G_N \Pi^{-1}$ . Its projection onto  $YF$  is

$$P_N^\theta = Y G_{N,F}^{-1} Y^* \tilde{G}_N = \Pi I G_{N,F}^{-1} I^* G_N \Pi^{-1}. \quad (\text{SC23})$$

Direct multiplication proves its idempotence and its  $\tilde{G}_N$ -self-adjointness. With  $z_N^\theta = (I_q - P_N^\theta) \Pi e_0$ , its squared target norm is the literal theta Schur complement

$$\|z_N^\theta\|_{\tilde{G}_N}^2 = e_0^* G_N e_0 - e_0^* G_N I G_{N,F}^{-1} I^* G_N e_0 > 0. \quad (\text{SC24})$$

The right side is positive because it is the norm of the nonzero  $G_N$ -orthogonal projection of  $e_0$  away from  $I(F)$ . For this moving target metric  $Y^* \tilde{G}_N Y = G_{N,F}$  is constant. Its differentiated metric terms are explicitly

$$\partial_t \tilde{G}_N = \Pi^{-*} G_N (A + t\mathcal{R}) \Pi^{-1} / u, \quad \partial_t \tilde{G}_N = \Pi^{-*} (A + t\mathcal{R})^* G_N \Pi^{-1} / \bar{u}. \quad (\text{SC25})$$

They cancel  $Y^* \tilde{G}_N Y'$  and its conjugate derivative in the product rule. Thus the constant pulled-back theta Gram and the nonzero curvature (SC16) are connected by the exact metric congruence (SC23)–(SC25), with every term retained.

For  $F = 0$ , use the empty Gram determinant 1, empty trace 0 and the zero normal map. For  $F = E$ ,  $P = I_q$  and  $N = 0$  identically, so (SC15) is zero: the full determinant logarithm in (SC5) is the real part of a holomorphic function plus its stated constant. The strictly positive coefficient in (SC16) belongs precisely to  $0 < p < q$ . This curvature and the original theta comparison are fully calculated observables; no uniform arithmetic upper bound is asserted by these finite-dimensional identities.

## 2.6 The exact derived base change at a collision

The trace and the positive-degree diagonal groups in the delivered SGA note have a further explicit specialization map. Retain the whole one-parameter algebra over  $T = \mathbb{C}[t]$ :

$$\mathcal{E} = T[S]/(\chi(S) - t) \simeq \mathbb{C}[S], \quad t \mapsto \chi(S), \quad M = \mathcal{E}/\chi' \mathcal{E}. \quad (\text{SC26})$$

The isomorphism sends  $[P(t, S)]$  to  $P(\chi(S), S)$ ; its inverse sends  $S$  to its original class. Monic division proves that  $\mathcal{E}$  is free of rank  $q$  over  $T$ . Since  $\chi'$  is a nonzero polynomial over  $\mathbb{C}$ , multiplication by  $\chi'$  is injective on  $\mathcal{E} \simeq \mathbb{C}[S]$ . Consequently

$$0 \longrightarrow \mathcal{E} \xrightarrow{\chi'} \mathcal{E} \longrightarrow M \longrightarrow 0 \quad (\text{SC27})$$

is a free  $T$ -resolution of  $M$ . The diagonal resolution gives  $HH_0(\mathcal{E}/T) = \mathcal{E}$ ,  $HH_{2a+1}(\mathcal{E}/T) = M$  and  $HH_{2a+2}(\mathcal{E}/T) = 0$  for  $a \geq 0$ . For clarity, that resolution is free over  $\mathcal{B} = \mathcal{E} \otimes_T \mathcal{E}$  with alternating maps  $X - Y$  and  $\mathcal{C}_H = (H(X) - H(Y))/(X - Y)$ , followed by the multiplication augmentation. Over  $(T[Y]/H(Y))[X]$  both factors are monic and therefore non-zero-divisors; their product is  $H(X)$  modulo  $H(Y)$ . Cancellation proves each alternating kernel equals the preceding image. Tensoring by the diagonal sets  $X = Y$  and gives the maps  $0, \chi', 0, \chi', \dots$ , proving the stated groups.

Fix  $t_0 \in \mathbb{C}$ , put  $\kappa = T/(t - t_0)$  and  $E_{t_0} = \mathbb{C}[S]/(\chi - t_0)$ . Tensoring (SC27) gives the complete kernel-and-cokernel sequence

$$\begin{aligned} 0 \longrightarrow \mathrm{Tor}_1^T(M, \kappa) &\longrightarrow E_{t_0} \xrightarrow{\chi'} E_{t_0} \\ &\longrightarrow M \otimes_T \kappa \longrightarrow 0. \end{aligned} \quad (\text{SC28})$$

It computes the actual specialized diagonal groups:

$$\begin{aligned} HH_{2a+1}(E_{t_0}/\mathbb{C}) &= M \otimes_T \kappa = E_{t_0}/(\chi'), \\ HH_{2a+2}(E_{t_0}/\mathbb{C}) &= \mathrm{Tor}_1^T(M, \kappa) = \mathrm{ann}_{E_{t_0}}(\chi'). \end{aligned} \quad a \geq 0. \quad (\text{SC29})$$

Every arrow is induced by the same polynomial multiplication, so these are equivariant maps for the original  $S$ -action.

There is also a representative-level description of the Tor map. Resolve  $\kappa$  by  $[T \xrightarrow{t-t_0} T]$  to identify  $\mathrm{Tor}_1^T(M, \kappa)$  with the kernel of  $t - t_0$  on  $M$ . If  $[v]_{\chi'}$  belongs to this kernel, write

$$(\chi - t_0)v = \chi'w. \quad [v]_{\chi'} \longmapsto [w]_{\chi-t_0}. \quad (\text{SC30})$$

Replacing  $v$  by  $v + \chi'a$  replaces  $w$  by  $w + (\chi - t_0)a$ , so the map is well-defined. Its image is annihilated by  $\chi'$ . If its image is zero,  $w = (\chi - t_0)a$  and cancellation of the nonzero polynomial  $\chi - t_0$  gives  $v = \chi'a$ , proving injectivity. Conversely an annihilated class  $[w]_{\chi-t_0}$  satisfies  $\chi'w = (\chi - t_0)v$  for a polynomial  $v$ , giving its preimage. These operations commute with multiplication by  $S$ .

At a root  $\lambda$  of multiplicity  $m$  in  $\chi - t_0$ , its local algebra is  $\mathbb{C}[x]/(x^m)$ ,  $x = S - \lambda$ , and  $\chi' = mx^{m-1}a(x)$  with the original invertible local factor  $a$  (this equality is in the local algebra: before reduction, the derivative of  $x^m a(x)$  also contains the term  $x^m a'(x)$ ). The odd group has local module  $\mathbb{C}[x]/(x^{m-1})$  and the even group has local module  $(x)/(x^m)$ ; both have dimension  $m - 1$ , with their full induced  $S = \lambda + x$  action. The map  $[b]_{x^{m-1}} \mapsto [xb]_{x^m}$  is an explicit module isomorphism between these local modules, using the stated local coordinate. For  $m = 1$  both groups vanish. Thus the zero even group of the family and the nonzero even group of a collision fibre are related by the exact Tor term (SC28), not by omitting specialization data. The coefficient maps in the residue construction commute with base change; this calculation retains the additional derived term required when their kernels are specialized. Lifting every arrow at its original Split-Zero support keeps its receiving supported zero and external absence distinct as in (SC14).

## 2.7 The exact neighboring-minor identity and its theta factors

The ideal basis in (SC9) supplies three actual polynomial inclusions. Write their increasing-power coefficient matrices as

$$J_g = [g, Sg, \dots, S^{p-1}g], \quad J_- = [g, Sg, \dots, S^{p-2}g], \quad J_+ = [J_g, e_0]. \quad (\text{SC31})$$

For  $p = 1$ ,  $J_-$  has zero columns. All three are injective on their stated coefficient spaces; the first two assertions follow from their distinct monic leading degrees, and the last from  $e_0 \notin I(F)$ . Let

$C = (I^*I)^{-1}I^*J_g$ . Since the columns of  $J_g$  lie in  $\text{im } I$ , this satisfies  $IC = J_g$ ; it is invertible because both matrices are bases of that same subspace. The exact frame equations are

$$H_g(t) := (\Pi J_g)^* \Pi J_g = C^* H(t) C, \quad LC = (0, \dots, 0, 1), \quad h_g(t) = |\det C|^2 \det H(t), \quad (\text{SC32})$$

where  $h_J(t) = \det((\Pi J)^* \Pi J)$  for each of the three inclusions. The scalar  $|\det C|^2$  has been retained; it is independent of  $t$ , so it has zero logarithmic derivatives.

The Gram of  $\Pi J_+$  has block form

$$\begin{pmatrix} H_g & (\Pi J_g)^* \Pi e_0 \\ (\Pi e_0)^* \Pi J_g & \|\Pi e_0\|^2 \end{pmatrix}. \quad (\text{SC33})$$

Eliminate its off-diagonal blocks using triangular matrices of determinant one. Its lower-right Schur complement is the squared norm of the orthogonal projection of  $\Pi e_0$  away from  $\text{im } \Pi J_g$ , which is  $\|z(t)\|^2$  by (SC11). Thus  $h_+ = h_g \|z\|^2$ . The last diagonal entry of  $H_g^{-1}$ , computed by the adjugate formula, is its upper-left principal minor divided by  $h_g$ , namely  $h_-/h_g$ . By (SC32) that entry is exactly  $LH^{-1}L^*$ . Formula (SC16) becomes

$$\partial_t \partial_t \log h_g(t) = \frac{|t|^2}{|u|^2} \frac{h_+(t)h_-(t)}{h_g(t)^2}. \quad (\text{SC34})$$

This identity retains all three original polynomial maps. It is the precise neighboring-minor equation joining the period calculation to the source-volume and Toda calculations; it does not identify differently ordered or differently weighted families.

For each same inclusion  $J = J_-, J_g, J_+$ , define its original theta volume and its positive comparison determinant by

$$v_{N,J} = \det(J^* G_N J), \quad b_{N,J}(t) = \det((J^* G_N J)^{-1} J^* \Pi(t)^* \Pi(t) J). \quad (\text{SC35})$$

Injectivity and positivity show that both numbers are positive. For the empty inclusion both are 1. Taking the determinant of the literal product gives  $h_J = v_{N,J} b_{N,J}$ , and therefore

$$c_F(t) = \frac{1}{|u|^2} \frac{v_{N,J_+} v_{N,J_-}}{v_{N,J_g}^2} \frac{b_{N,J_+}(t) b_{N,J_-}(t)}{b_{N,J_g}(t)^2}. \quad (\text{SC36})$$

Every original theta mass and coefficient inclusion remains in the  $v$  factors; every period comparison remains in the  $b$  factors. This proves the metric dictionary required to use (SC34) within that original source programme.

For  $p = q - 1$ , the divisor  $g$  is monic linear. The square matrix  $J_+ = [g, Sg, \dots, S^{q-2}g, e_0]$  has determinant  $(-1)^{q-1}$ : expand along its last column, and the remaining minor on rows  $1, \dots, q - 1$  is upper triangular with every diagonal entry 1. Thus, with that sign retained,

$$\det(\Pi J_+) = (-1)^{q-1} \det \Pi, \quad h_+(t) = (2\pi|u|)^q \exp\left(2 \operatorname{Re} \frac{\operatorname{Tr} \Phi_t(A + t\mathcal{R})}{u}\right). \quad (\text{SC37})$$

The top minor in (SC34) is therefore the explicitly evaluated full determinant in this codimension-one case. The remaining two minors and their exact theta factors are still those in (SC31)–(SC36).

### 3 The period curvature of the retained critical-comparison kernel

This section joins the original theta representatives, the critical-line comparison of the Hochschild note (H16)–(H25), and the period curvature (SC1)–(SC37). The single-packet identification is proved first, including its full arithmetic unit. We then calculate the factorwise critical comparison on every cyclic tensor packet. All tensor targets below use the completed projective tensor product of the programme’s locally convex spaces; its finite surviving image is proved injective by a continuous retraction, without an assertion of tensor exactness for arbitrary spaces.

The proof-bearing comparison source is *The original theta complex inside the Hochschild trace realization*, (H1)–(H25), at the retained PR27 revision `a4a87844c257485ae6d669f5b3eb6d96b9e8b8d8`; its original analytic constructions are retained in the cumulative source. The arithmetic section and unit maps are (A7)–(A13), (CT.1)–(CT.5), and (TVB.1)–(TVB.3). The calculations here prove the additional maps between those specified objects.

#### 3.1 The single-packet polynomial and its complete arithmetic unit

Retain the original nonempty packet, with every selected zero’s complete order,

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad g(s) = 2\xi(s), \quad v_h(s) = g(s)/h(s), \quad d = \deg h. \quad (\text{PC1})$$

The packet is the same one used by the original source, including its reflection symmetry when that symmetry is used in its metric. Put  $E_h = \mathbb{C}[s]/(h)$ , and let  $j_h$  mean the full Hermite remainder. Its local coordinate at  $\rho$  is  $x = s - \rho$ , with basis  $1, x, \dots, x^{m_\rho-1}$ . Thus

$$v_{h,\rho} = \sum_{a=0}^{m_\rho-1} \frac{v_h^{(a)}(\rho)}{a!} x^a, \quad v_h = j_h(v_h), \quad \varepsilon_h = v_h^{-1}. \quad (\text{PC2})$$

This inverse exists with all its coefficients retained. Indeed write  $h(s) = (s - \rho)^{m_\rho} h_\rho(s)$  and  $g(s) = (s - \rho)^{m_\rho} g_\rho(s)$ . Both remaining factors are nonzero at  $\rho$ , so the constant coefficient  $a_0 = g_\rho(\rho)/h_\rho(\rho)$  is nonzero. If the coefficients of  $v_{h,\rho}$  are  $a_a$ , its inverse coefficients satisfy

$$b_0 = a_0^{-1}, \quad b_n = -a_0^{-1} \sum_{a=1}^n a_a b_{n-a} \quad (1 \leq n < m_\rho). \quad (\text{PC3})$$

Multiplication makes the constant coefficient one and each other coefficient through degree  $m_\rho - 1$  zero. The coprime polynomial factors of  $h$  give the Chinese-remainder isomorphism; assembling these inverses proves (PC2). The notation  $j_h(h/g)$  for  $\varepsilon_h$  uses its holomorphic germ at every selected zero; no global entire reciprocal is asserted.

For  $k = 1$ , the sum variable is literally  $S = s_1$ . The isomorphism  $\mathbb{C}[S] \rightarrow \mathbb{C}[s_1]$ ,  $S \mapsto s_1$ , pulls  $(h(s_1))$  back to  $(h(S))$ . The monic annihilator defining the cyclic packet is consequently

$$\begin{aligned} \chi_{h,1}(S) &= h(S), \quad E = \mathbb{C}[S]/(h(S)), \quad A = M_S, \quad q = d, \\ \iota_1 : E &\xrightarrow{\sim} E_h, \quad [P(S)] \mapsto [P(s)], \quad \eta_h = M_{v_h} \iota_1. \end{aligned} \quad (\text{PC4})$$

The equality includes every exponent  $m_\rho$ . The inverse of  $\eta_h$  is  $\iota_1^{-1} M_{\varepsilon_h}$ . It intertwines the two original multiplication actions because multiplication by the unit commutes with  $s$ . Its actual multiplicative rule is  $\eta_h(x)\eta_h(y) = v_h\eta_h(xy)$  and  $\eta_h(1) = v_h$ . Thus its source and target polynomial algebras and its module-map type remain explicit.

Use the original spaces and maps

$$\begin{aligned}
V &= \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = \int_{\mathbb{R}} \phi = 0\}, \quad D = -x\partial_x, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \\
\mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for all } b \in \mathbb{Z}, j \geq 0\}, \quad Q = \mathcal{B}/\Theta V, \quad \mathbf{q} : \mathcal{B} \rightarrow Q, \\
\mathcal{M}F(s) &= \int_0^\infty F(x) x^{s-1} dx, \quad \phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M}F_h = v_h, \quad h(D)F_h = \Theta\phi_*.
\end{aligned} \tag{PC5}$$

The source construction of  $F_h$  is the iterated original Euler inverse  $S_\rho F = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy$  on the vanishing Mellin fibre, as proved in (A5)–(A7) and (TVB.1). Let  $\mathcal{T}_h P = P(D)F_h$ ,  $\pi_h P = [P]_h$ , and  $J_h = j_h \mathcal{M} : \mathcal{B} \rightarrow E_h$ , with the retained  $\sigma_h : E_h \rightarrow Q$ . The exact source relations are

$$\mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h = \eta_h \pi_h, \quad \mathbf{q} \mathcal{T}_h = \sigma_h \eta_h \pi_h. \tag{PC6}$$

Here and below  $\pi_h$  uses the variable identification (PC4). The first identity follows from  $h(D)F_h = \Theta\phi_*$  and commutation with  $D$ . Integration by parts gives  $\mathcal{M}\mathcal{T}_h P = v_h P$ , proving the second identity. For the third, the retained section in (A10) is  $\sigma_h = \mathbf{q} R_{\text{ref}}$ , with  $R_{\text{ref}} u = \mathcal{T}_h \text{rem}_h(\varepsilon_h u)$ . Write  $P = P_0 + hT$ ,  $\deg P_0 < d$ . Then  $R_{\text{ref}} \eta_h[P] = \mathcal{T}_h P_0$ , and the difference from  $\mathcal{T}_h P$  is the displayed theta boundary  $\Theta(T(D)\phi_*)$ . This proves the precise equality with that section, rather than identifying two sections merely from equal jets. Since  $J_h \Theta = 0$ , its descended map is  $\bar{J}_h : Q \rightarrow E_h$ , defined by  $\bar{J}_h \mathbf{q} = J_h$ ; the section identity is  $\bar{J}_h \sigma_h = I_{E_h}$ . Also  $D\mathcal{T}_h P = \mathcal{T}_h(SP)$  and (PC6) imply  $D_Q \sigma_h \eta_h = \sigma_h \eta_h A$ . Surjectivity of  $\eta_h$  therefore proves  $D_Q \sigma_h = \sigma_h M_s$  with its exact domains.

For each admitted original polynomial degree  $N$ , retain its actual remainder matrix  $B_N$ , source Gram  $O_N$ , and least representative

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad r_N = \mathcal{T}_h R_N. \tag{PC7}$$

The source Gram is for the original density  $|v_h(1/2 + iy)|^2 dy / (2\pi)$  and the original monic coordinates. Since  $B_N R_N = I$ , (PC6) gives

$$j_E := \eta_h^{-1} J_h : \mathcal{B} \rightarrow E, \quad j_E r_N = I, \quad J_h r_N = \eta_h, \quad \mathbf{q} r_N = \sigma_h \eta_h, \quad j_E D = A j_E. \tag{PC8}$$

The last identity uses  $J_h D = M_s J_h$  and the intertwining in (PC4). In particular the coefficient observer in (SC7) is  $j_E$ . In full Mellin-jet coordinates the same Gram is  $\eta_h^{-*} G_N \eta_h^{-1}$ ; multiplying back by  $\eta_h^*$  and  $\eta_h$  recovers exactly  $G_N$ . The unit is present in both the source map and the coordinate congruence.

### 3.2 The weighted critical comparison and every recovered jet

Reserve  $y \in \mathbb{R}$  for the critical-line coordinate, distinct from the complex period parameter  $t$ . The retained continuous map is

$$Q_{\text{crit}} = \mathcal{S}(\mathbb{R}) / \overline{\zeta(1/2 + iy) \mathcal{S}(\mathbb{R})}, \quad \Gamma[F] = [\mathcal{M}F(1/2 + iy)], \quad M = M_{1/2+iy}, \quad \Gamma D_Q = M \Gamma. \tag{PC9}$$

It is induced by the plus-sign Mellin restriction: the theta image restricts to  $2\zeta(1/2 + iy) \mathcal{M}\phi(1/2 + iy)$ . The continuity and strong-Schwartz domains are those proved in (H16)–(H20); continuity also makes it factor through the closed theta quotient. Let  $Z_c = \{\rho \in Z : \Re \rho = 1/2\}$  and  $Z_o = Z \setminus Z_c$ . Retain the complete factors

$$h_c = \prod_{\rho \in Z_c} (S - \rho)^{m_\rho}, \quad h_o = \prod_{\rho \in Z_o} (S - \rho)^{m_\rho}, \quad h = h_c h_o. \tag{PC10}$$

An empty product is 1. Chinese remainders identify  $E = E_c \oplus E_o$ , where  $E_c$  is the ideal  $(h_o)/(h)$  and  $E_o$  is the ideal  $(h_c)/(h)$ , with projections  $\pi_c, \pi_o$ . The notation also retains the isomorphisms to the products of their local primary algebras, including their nilpotent coordinates.

Define the actual weighted comparison

$$\tilde{\sigma}_h = \sigma_h \eta_h : E \rightarrow Q, \quad \mathcal{C}_h = \Gamma \tilde{\sigma}_h : E \rightarrow Q_{\text{crit}}. \quad (\text{PC11})$$

Then

$$\boxed{\ker \mathcal{C}_h = E_o, \quad \mathcal{C}_h A = M \mathcal{C}_h.} \quad (\text{PC12})$$

To prove the kernel, first take an off-critical root  $\rho$ . The reciprocal  $(1/2 + iy - \rho)^{-1}$  and all its derivatives have polynomial bounds on the real axis. Multiplication by it preserves Schwartz space, the zeta ideal and its closure, and is inverse to  $M - \rho$  on  $Q_{\text{crit}}$ . On a primary block  $(A - \rho)^{m_\rho} = 0$ , intertwining therefore forces the whole block to zero. Multiplication by  $v_h$  preserves each CRT block and does not change this conclusion.

For a critical root  $\rho = 1/2 + i\gamma$  retain the evaluation

$$\Xi_\rho[f] = \sum_{j=0}^{m_\rho-1} \frac{i^{-j}}{j!} f^{(j)}(\gamma) x^j \in \mathbb{C}[x]/(x^{m_\rho}), \quad x = S - \rho. \quad (\text{PC13})$$

Every derivative used here vanishes on the zeta ideal and, by continuity, on its closure. The chain rule supplies  $i^j$  when differentiating  $\mathcal{M}F(1/2 + iy)$ , so (PC6) and  $\bar{J}_h \sigma_h = I$  give the full formula

$$\Xi_\rho \mathcal{C}_h[P] = v_{h,\rho}[P]_\rho = \sum_{j < m_\rho} \sum_{a=0}^j \frac{v_h^{(a)}(\rho)}{a!} \frac{P^{(j-a)}(\rho)}{(j-a)!} x^j. \quad (\text{PC14})$$

In particular the critical observer recovers the actual unit on  $e_0 = [1]$ , rather than the constant one. Assemble the  $\Xi_\rho$  by the retained CRT map and then  $\iota_1^{-1}$  into  $\Xi_c : Q_{\text{crit}} \rightarrow E_c$ . Put  $v_c = \pi_c \iota_1^{-1}(v_h)$  and  $\varepsilon_c = \pi_c \iota_1^{-1}(\varepsilon_h)$ ; these are inverse units in the ideal algebra  $E_c$ , whose identity is  $e_c = \pi_c(e_0)$ . Multiplication by this inverse defines the continuous map

$$\mathcal{R}_c := M_{\varepsilon_c} \Xi_c, \quad \mathcal{R}_c \mathcal{C}_h = \pi_c. \quad (\text{PC15})$$

This is a left inverse on the entire critical block, proving (PC12). For an empty critical packet both maps have zero-dimensional target. When  $E_c \neq 0$ , (PC14) proves  $\mathcal{C}_h(e_0) \neq 0$ . The finite image  $W_h = \mathcal{C}_h(E_c)$  is preserved by  $M$ , and  $\mathcal{C}_h|_{E_c}$  and  $\mathcal{R}_c|_{W_h}$  are inverse maps.

### 3.3 The comparison kernel is a holomorphic period subbundle

Use exactly the polynomial  $\chi = h$  in (SC1)–(SC6), its potential

$$\Phi_t(S) = \frac{S^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS, \quad h(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad u \in \mathbb{C}^*. \quad (\text{PC16})$$

Fix the original argument of  $u$  and rays  $\theta_j = (\arg u + \pi + 2\pi j)/(q+1)$ , with  $\Gamma_j = -\ell_0^{\text{ray}} + \ell_j^{\text{ray}}$  and  $\Pi_{jb}(t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS$ . The original monomial factors are

$$(\Pi_{\text{mon}})_{j,b} = \frac{((q+1)|u|)^{(b+1)/(q+1)}}{q+1} \Gamma\left(\frac{b+1}{q+1}\right) e^{i(b+1)\theta_0} (e^{2\pi i j(b+1)/(q+1)} - 1). \quad (\text{PC17})$$

They have nonzero determinant: a column dependence of their last matrix gives a polynomial of degree at most  $q$  vanishing at all  $q + 1$  roots of unity. Its coefficients must be zero. Along any compact coefficient path the negative leading ray term dominates the lower-degree terms by an integrable bound  $Cr^L \exp(-r^{q+1}/(2(q+1)|u|))$ . Differentiation under the integral is therefore valid. The original polynomial reduction  $uP' + (h - t)P$  has leading degree  $\deg P + q$  and the same leading coefficient, so it has a unique remainder below degree  $q$ . Its integral vanishes by the exact derivative and cancellation of the two origin values. Coefficient variation thus gives  $\Pi' = \Pi B$  along a path, and the adjugate determinant identity gives  $\det \Pi(s) = \det \Pi(0) \exp(\int \text{Tr } B)$ . This proves invertibility at every  $t$ , including all collisions.

In the unchanged coefficient basis set

$$\ell([P]) = [S^{q-1}] \text{rem}_h P, \quad \mathcal{R} = e_0 \ell, \quad A_t = A + t\mathcal{R}, \quad \Pi'(t) = -\Pi(t)A_t/u. \quad (\text{PC18})$$

The last equality follows by differentiating in  $t$ : for the final column the sole reduction is  $[h - t] = 0$ , obtained from the differential of the constant polynomial one. Define maps on the period target, with range still in the original critical quotient,

$$\mathcal{C}_{h,t} = \mathcal{C}_h \Pi(t)^{-1}, \quad A_t^{\text{per}} = \Pi(t)A_t \Pi(t)^{-1}, \quad \mathcal{K}_{h,t} = \ker \mathcal{C}_{h,t} = \Pi(t)E_o. \quad (\text{PC19})$$

The equality of kernels follows from invertibility and (PC12). The inverse matrix is holomorphic, so this is a holomorphic kernel bundle of constant rank, with the indicated global frame from  $E_o$ . Its quotient is the fixed finite image  $W_h$ ; more explicitly  $\mathcal{C}_{h,t} \Pi(t)|_{E_c} = \mathcal{C}_h|_{E_c}$  and  $\Pi(t)|_{E_c} \mathcal{R}_c$  is a right inverse on  $W_h$ .

The exact action defect is

$$\boxed{\mathcal{C}_{h,t} A_t^{\text{per}} - M \mathcal{C}_{h,t} = t \mathcal{C}_h(e_0) \ell \Pi(t)^{-1}}. \quad (\text{PC20})$$

Indeed multiplying the first term gives  $\mathcal{C}_h(A + t\mathcal{R})\Pi^{-1}$ , and the part containing  $A$  cancels  $M \mathcal{C}_h \Pi^{-1}$  by (PC12). This proves the formula before restricting its domain.

Put  $F = E_o$ ,  $p = \dim F$ . If  $0 < p < q$ , it is the ideal  $(h_c)/(h)$ , with ordered basis  $[h_c], [Sh_c], \dots, [S^{p-1}h_c]$ . The last member is monic of degree  $q - 1$ , so  $\ell|_F$  is onto with kernel dimension  $p - 1$ . An ideal containing  $e_0$  would be all of  $E$ , hence  $e_0 \notin F$ . It follows from (PC14) and (PC20) that, for  $t \neq 0$ , the restriction of this action defect to  $\mathcal{K}_{h,t}$  has

$$\text{im} = \mathbb{C} \mathcal{C}_h(e_0), \quad \ker = \Pi(t)(F \cap \ker \ell), \quad \text{rank} = 1. \quad (\text{PC21})$$

At  $t = 0$  it is zero. If  $F = 0$  its restricted domain is zero. If  $F = E$ , the comparison  $\mathcal{C}_h$  is zero and every displayed defect is zero. These statements cover all actual sector possibilities without asserting that an off-critical zero exists.

### 3.4 The full curvature and both original metric observations

Take any specified coefficient inclusion  $I : F \hookrightarrow E$ ; put  $L = \ell I$ , and for  $0 < p < q$  define

$$Y = \Pi I, \quad H = Y^* Y, \quad P = Y H^{-1} Y^*, \quad N = I_q - P, \quad z = N \Pi e_0. \quad (\text{PC22})$$

This is exactly the kernel bundle (PC19), measured in the fixed period-coordinate Gram  $I_q$ . The Gram is positive because  $Y$  is injective. Multiplication proves that  $P$  is its orthogonal projection. Since  $e_0 \notin F$  and  $\Pi$  is invertible,  $z(t) \neq 0$  for every  $t$ . Original invariance  $AI = IA_F$  gives

$$Y' = -Y A_F/u - (t/u) \Pi e_0 L, \quad N Y' = -(t/u) z L, \quad N(0) Y''(0) = -z(0) L/u. \quad (\text{PC23})$$

The final equation follows by differentiating (PC18); the term  $\Pi(0)A^2I/u^2$  lies inside the kernel frame and is removed by  $N(0)$ . Thus the first normal derivative vanishes at zero and the normal acceleration has rank one, with its original  $u$  factor.

For  $t = x + iy_t$  use  $\partial_t = (\partial_x - i\partial_{y_t})/2$  and its conjugate. Holomorphicity gives  $\partial_t H = Y^*Y'$ ,  $\partial_{\bar{t}} H = Y'^*Y$ , and  $\partial_{\bar{t}}\partial_t H = Y'^*Y'$ . Differentiate  $H^{-1}H = I$  and the determinant to obtain

$$\begin{aligned}\partial_{\bar{t}}\partial_t \log \det H &= \text{Tr}(H^{-1}Y'^*Y' - H^{-1}Y'^*YH^{-1}Y^*Y') \\ &= \text{Tr}(H^{-1}Y'^*NY') = \frac{|t|^2}{|u|^2} \|z\|^2 LH^{-1}L^*.\end{aligned}\tag{PC24}$$

Every factor in the last coefficient is strictly positive except the explicit  $|t|^2$ :  $z \neq 0$ ,  $L \neq 0$  and  $H^{-1} > 0$ . The coefficient is real analytic, so the curvature vanishes at zero to exactly quadratic order. In the determinant-dual metric  $(\det H)^{-1}$  its first Chern form is  $\frac{i}{2\pi} \frac{|t|^2}{|u|^2} \|z\|^2 LH^{-1}L^* dt \wedge d\bar{t}$ , with  $i dt \wedge d\bar{t} = 2dx \wedge dy_t$ . This proves that the SC curvature is the curvature of this actual critical-comparison kernel in the specified fixed period metric.

The original theta Gram remains separately and explicitly present. Let  $G_{N,F} = I^*G_N I$ , and set

$$B_{N,F} = G_{N,F}^{-1}H, \quad \tilde{G}_N = \Pi^{-*}G_N\Pi^{-1}, \quad P_N^\theta = \Pi I G_{N,F}^{-1} I^* G_N \Pi^{-1}.\tag{PC25}$$

Thus  $H = G_{N,F}B_{N,F}$  and  $Y^*\tilde{G}_N Y = G_{N,F}$ . Both identities follow by direct multiplication. The squared normal norm of  $\Pi e_0$  for the moving target metric is exactly

$$e_0^* G_N e_0 - e_0^* G_N I G_{N,F}^{-1} I^* G_N e_0 > 0.\tag{PC26}$$

It is positive because it is the squared norm of the nonzero  $G_N$ -orthogonal component of  $e_0$  outside  $F$ . From (PC18),  $\partial_t \Pi^{-1} = A_t \Pi^{-1}/u$ , so

$$\partial_t \tilde{G}_N = \Pi^{-*} G_N A_t \Pi^{-1}/u, \quad \partial_{\bar{t}} \tilde{G}_N = \Pi^{-*} A_t^* G_N \Pi^{-1}/\bar{u}.\tag{PC27}$$

In the product rule for  $Y^*\tilde{G}_N Y$  these terms cancel the derivatives of  $Y$  and  $Y^*$  exactly. Hence its Gram and determinant are constant in the same holomorphic frame. Formula (PC24) and this cancellation are related by the full displayed metric congruence, including all original theta masses and unit factors. For  $F = 0$  the determinant is 1; for  $F = E$ ,  $N = 0$ . Both endpoint curvatures are therefore zero.

### 3.5 The exact completed tensor comparison and its collision jets

We now retain an arbitrary original integer  $k \geq 1$ . Set

$$B_k = E_h^{\otimes k} = \mathbb{C}[s_1, \dots, s_k]/(h(s_1), \dots, h(s_k)), \quad S = s_1 + \dots + s_k, \quad U_k = v_h^{\otimes k},\tag{PC28}$$

and retain the original cyclic algebra and its two distinct maps. The variable isomorphism between its tensor coefficient carriers is  $\iota_k = \iota_1^{\otimes k} : E^{\otimes k} \xrightarrow{\sim} B_k$ . Thus

$$E_k = \mathbb{C}[S]/(\chi_{h,k}), \quad A_k = M_S, \quad \alpha_k[P] = [P(S)] \in B_k^{\mathfrak{S}_k}, \quad \eta_k = M_{U_k} \alpha_k.\tag{PC29}$$

The map  $\alpha_k$  is unital;  $\eta_k$  is the original unit-weighted module injection. Define the fully typed comparison  $\bar{\alpha}_k = \iota_k^{-1} \alpha_k : E_k \rightarrow E^{\otimes k}$ . Then  $\eta_k = \eta_h^{\otimes k} \bar{\alpha}_k$ , since  $\eta_h = M_{v_h} \iota_1$ . Their source polynomial is exactly

$$\chi_{h,k}(S) = \prod_{\lambda} (S - \lambda)^{m_{\lambda}}, \quad m_{\lambda} = \max_{\substack{\rho_1 + \dots + \rho_k = \lambda \\ \rho_i \in Z}} \left( 1 + \sum_{i=1}^k (m_{\rho_i} - 1) \right).\tag{PC30}$$

All tuples are ordered in the full tensor algebra; equal sums are grouped by their literal equality. To verify the exponent, the local factor at a tuple has coordinates  $x_i = s_i - \rho_i$  and quotient by  $(x_i^{m_{\rho_i}})$ . The nilpotent sum  $x_1 + \cdots + x_k$  has highest nonzero power  $D = \sum_i (m_{\rho_i} - 1)$ : that power contains the monomial  $\prod_i x_i^{m_{\rho_i} - 1}$  with the nonzero coefficient  $D! / \prod_i (m_{\rho_i} - 1)!$ . Higher powers vanish. Taking the maximum over equal sums proves (PC30) and the injectivity of  $\alpha_k$ . The invariant unit vector already has that same annihilator, so passage to symmetric invariants does not alter it.

Let  $\chi_{c,k}$  be the same polynomial formed using only tuples from  $Z_c$ , with exponents  $r_\lambda$  given by the same maximum; an absent critical tuple at  $\lambda$  has  $r_\lambda = 0$ . When  $Z_c = \emptyset$ , define  $\chi_{c,k} = 1$ . Put  $q_k = \deg \chi_{h,k}$  and  $r_k = \deg \chi_{c,k}$ . The tuple inclusion proves  $r_\lambda \leq m_\lambda$  and hence  $\chi_{c,k} \mid \chi_{h,k}$ . The same nilpotence proof gives the exact injection

$$\alpha_{c,k} : \mathbb{C}[S]/(\chi_{c,k}) \hookrightarrow E_c^{\otimes k}, \quad [P] \mapsto P(A_c^{[k]})e_c^{\otimes k}, \quad \pi_c^{\otimes k} \bar{\alpha}_k = \alpha_{c,k} \pi_{\chi_{c,k}}. \quad (\text{PC31})$$

Here  $A_c = A|E_c$  and  $A_c^{[k]}$  is its sum on the ordered tensor factors. The identity vector is the displayed  $e_c^{\otimes k}$ , so the codomain is precisely the original CRT ideal tensor. For an empty critical factor both the source and target of  $\alpha_{c,k}$  are zero. This identity keeps the full source polynomial  $\chi_{h,k}$  and the actual quotient map to its critical tuple polynomial.

The target is the completed projective tensor space  $Q_{\text{crit}}^{\widehat{\otimes} \pi^k}$ . There is no loss of the finite critical image on completion. In fact define  $i_c = \mathcal{C}_h|E_c$  and use the continuous map  $\mathcal{R}_c$  in (PC15). Their tensor maps extend continuously to the projective completions, and on pure tensors their composition is identity. Pure tensors span the finite-dimensional  $E_c^{\otimes k}$ , which is already complete, so

$$\mathcal{R}_c^{\widehat{\otimes} k} i_c^{\widehat{\otimes} k} = I_{E_c^{\otimes k}}. \quad (\text{PC32})$$

This proves injectivity and a continuous retraction onto the finite image inside the completed target. Its associated continuous idempotent has that image as its kernel-complement, so the image is closed in the Hausdorff completed space. This argument uses the displayed maps and does not assume preservation of arbitrary exact sequences by completion.

The actual tensor comparison, with its original weighted inclusion, is

$$\mathcal{C}_k = \Gamma^{\widehat{\otimes} k} \sigma_h^{\widehat{\otimes} k} \eta_k = \mathcal{C}_h^{\widehat{\otimes} k} \bar{\alpha}_k : E_k \rightarrow Q_{\text{crit}}^{\widehat{\otimes} \pi^k}. \quad (\text{PC33})$$

Here  $\sigma_h^{\widehat{\otimes} k} \eta_k$  is the existing arithmetic cohomology inclusion. Applying the raw critical evaluations gives  $\Xi_c^{\widehat{\otimes} k} \mathcal{C}_k = (M_{v_c})^{\otimes k} \alpha_{c,k} \pi_{\chi_{c,k}}$ . Applying their inverses (PC32) instead gives  $\alpha_{c,k} \pi_{\chi_{c,k}}$ . Every derivative coordinate has its product phase  $i^{-\sum j_i} / \prod j_i!$  and the full tensor of the local units (PC14). The injectivity in (PC31)–(PC32) therefore proves the exact kernel and image dimension

$$\boxed{F_k := \ker \mathcal{C}_k = (\chi_{c,k})/(\chi_{h,k}), \quad \dim F_k = q_k - r_k, \quad \dim \text{im } \mathcal{C}_k = r_k.} \quad (\text{PC34})$$

To display all collision data, at an ambient sum  $\lambda$  write  $x = S - \lambda$  and  $\chi_{c,k}(S) = x^{r_\lambda} a_\lambda(x)$  with  $a_\lambda(0) \neq 0$ . The kernel in that original primary factor is the image of the explicit map

$$\mathbb{C}[x]/(x^{m_\lambda - r_\lambda}) \longrightarrow \mathbb{C}[x]/(x^{m_\lambda}), \quad [b] \longmapsto [x^{r_\lambda} a_\lambda(x) b]. \quad (\text{PC35})$$

It is injective: multiply by the recursively computed inverse of  $a_\lambda$  and cancel the displayed power of  $x$ . It is onto the local ideal by its definition. Thus all  $m_\lambda - r_\lambda$  kernel jets remain, with action  $S = \lambda + x$  and the local unit  $a_\lambda$  retained. If  $\Re \lambda \neq k/2$ , no all-critical tuple has that sum, so  $r_\lambda = 0$  and the entire block belongs to the kernel. At a sum with real part  $k/2$ , the same exact formula still applies: an absent critical tuple kills the whole block, and a coinciding critical tuple retains precisely

the quotient of length  $r_\lambda$  while the additional jets lie in (PC35). This is the complete relationship between factorwise critical selection and selection by the real part of the sum.

The original sum action also descends with its exact type. Multiplication by  $1/2 + iy$  is continuous on the one-factor quotient and preserves its relation space. On the completed tensor target define the continuous operator

$$M^{[k]} = \sum_{a=1}^k I^{\widehat{\otimes}(a-1)} \widehat{\otimes} M \widehat{\otimes} I^{\widehat{\otimes}(k-a)}. \quad (\text{PC36})$$

On pure tensors this is the sum of the original  $k$  multipliers, and continuity extends it to the completed target. Since each  $\eta_h$  commutes with the original one-factor action, (PC12) gives  $\mathcal{C}_k A_k = M^{[k]} \mathcal{C}_k$  on the cyclic image. At source level, the finite multinomial expansion of  $P(D_1 + \dots + D_k) F_h^{\otimes k}$  and (PC6) give  $\mathfrak{q}^{\widehat{\otimes} k} \mathcal{V}_k P = \sigma_h^{\widehat{\otimes} k} \eta_k[P]$ . Thus both the completed comparison and its sum action are induced by the original weighted source, with no replacement of  $U_k$ .

### 3.6 The same kernel curvature in every original cyclic tensor packet

Form the original period matrix  $\Pi_k(t)$  of (SC1)–(SC6) with the literal polynomial  $\chi_{h,k}$ , its full increasing-power basis, the same specified  $u \neq 0$ , and its own degree  $q_k$  in the ray formula. Write  $e_{0,k} = [1]$ ,  $\ell_k[P] = [S^{q_k-1}] \text{rem}_{\chi_{h,k}} P$  and  $A_{k,t} = A_k + t e_{0,k} \ell_k$ . The preceding convergence, remainder and determinant argument applies to this polynomial with all its coefficients; hence  $\Pi'_k = -\Pi_k A_{k,t}/u$  and it is invertible at every  $t$ . Define

$$\mathcal{C}_{k,t} = \mathcal{C}_k \Pi_k^{-1}, \quad A_{k,t}^{\text{per}} = \Pi_k A_{k,t} \Pi_k^{-1}, \quad \mathcal{K}_{k,t} = \Pi_k F_k. \quad (\text{PC37})$$

Equations (PC34) and (PC36) prove

$$\ker \mathcal{C}_{k,t} = \mathcal{K}_{k,t}, \quad \mathcal{C}_{k,t} A_{k,t}^{\text{per}} - M^{[k]} \mathcal{C}_{k,t} = t \mathcal{C}_k (e_{0,k}) \ell_k \Pi_k^{-1}. \quad (\text{PC38})$$

If  $0 < r_k < q_k$ , then  $\mathcal{C}_k(e_{0,k}) \neq 0$  by (PC31)–(PC33). Put  $p_k = q_k - r_k$ . The kernel ideal has basis  $[\chi_{c,k}], \dots, [S^{p_k-1} \chi_{c,k}]$ , with the last member monic of degree  $q_k - 1$ . Consequently  $\ell_k|F_k$  is onto, its kernel has dimension  $p_k - 1$ , and  $e_{0,k} \notin F_k$ . For every  $t \neq 0$ , the restriction of (PC38) to  $\mathcal{K}_{k,t}$  has rank one, image  $\mathbb{C} \mathcal{C}_k(e_{0,k})$ , and full kernel  $\Pi_k(F_k \cap \ker \ell_k)$ . At zero it vanishes.

With any specified inclusion  $I_k : F_k \hookrightarrow E_k$ , put  $Y_k = \Pi_k I_k$ ,  $H_k = Y_k^* Y_k$ ,  $N_k = I - Y_k H_k^{-1} Y_k^*$ ,  $z_k = N_k \Pi_k e_{0,k}$  and  $L_k = \ell_k I_k$ . The calculations (PC23)–(PC24) now give

$$\partial_{\bar{t}} \partial_t \log \det H_k = \frac{|t|^2}{|u|^2} \|z_k\|^2 L_k H_k^{-1} L_k^*. \quad (\text{PC39})$$

Its coefficient is strictly positive for  $0 < r_k < q_k$ , and the normal acceleration at zero is  $-z_k(0) L_k / u$  with rank one. This calculation includes the collision-kernel jets of (PC35). The complete neighboring-minor form is obtained with

$$\begin{aligned} J_g &= [\chi_{c,k}, S \chi_{c,k}, \dots, S^{p_k-1} \chi_{c,k}], \quad J_- = [\chi_{c,k}, \dots, S^{p_k-2} \chi_{c,k}], \\ J_+ &= [J_g, e_{0,k}], \quad h_J = \det((\Pi_k J)^* \Pi_k J). \end{aligned} \quad (\text{PC40})$$

For  $p_k = 1$ ,  $J_-$  is empty and  $h_- = 1$ . The Schur complement of the last column gives  $h_+ = h_g \|z_k\|^2$ ; the adjugate formula for the last inverse-Gram entry gives  $L_k H_k^{-1} L_k^* = h_- / h_g$  after the exact constant frame congruence of (SC32). Therefore

$$\partial_{\bar{t}} \partial_t \log h_g = \frac{|t|^2}{|u|^2} \frac{h_+ h_-}{h_g^2}. \quad (\text{PC41})$$

For the original tensor-source Gram  $G_{N,k}$ , each of these same inclusions has factors  $v_{N,J} = \det(J^*G_{N,k}J)$  and  $b_{N,J} = \det((J^*G_{N,k}J)^{-1}J^*\Pi_k^*\Pi_kJ)$ . The literal product identity  $h_J = v_{N,J}b_{N,J}$  retains every theta mass, polynomial map and period factor in (PC41). The moving metric  $\Pi_k^{-*}G_{N,k}\Pi_k^{-1}$  has constant kernel Gram  $I_k^*G_{N,k}I_k$ , with the full derivative cancellation (PC27), replacing  $A_t$  by  $A_{k,t}$ . These are the two exact metric observations on the same critical-comparison kernel.

If  $Z_c = \emptyset$ , then  $r_k = 0$ ,  $F_k = E_k$  and  $\mathcal{C}_k = 0$ ; the full kernel bundle has zero curvature. If all packet zeros are critical,  $\chi_{c,k} = \chi_{h,k}$ ,  $F_k = 0$ , and the empty-kernel curvature is zero. If both sectors occur, an all-critical tuple proves  $r_k > 0$ , while a repeated off-critical root has sum  $k\rho$  of real part different from  $k/2$ , proving  $r_k < q_k$ . Thus the strict case in (PC39) has an exact sector criterion at every original  $k$ .

Finally each coefficient arrow in this section has its original Split-Zero lift:  $v^\bullet \mapsto (Tv)^\bullet$  and  $\tau \mapsto \tau$ , with the retained support label. The complete kernel (PC34), its smaller action-defect kernel in (PC38), and its cohomological source image remain separately represented. A vector killed by one of these comparisons lands at that receiving fibre's supported zero. The amplitude projection maps both that zero and external absence to ordinary zero, and the displayed split maps commute with that projection. This preserves the original support and the full arithmetic unit through the critical quotient, the completed tensor target, and the period curvature calculation.

## 4 The exact diamond of the circle and tensor-critical observations

This calculation uses the original packet  $h$ , the original cyclic polynomial  $\chi = \chi_{h,k}$ , and both actual observation maps. Its proof dependencies are the complete *Periodized theta-source recovery and arithmetic endpoint control*, (P1)–(P64), in `sources/web_periodized_source_delivery/Tau_Periodized_Source_Control/NOTE.tex`, and the complete period/critical-kernel bridge (PC1)–(PC41), in `work/period_critical_kernel_bridge_20260913.tex`. The former is the delivered source of SHA256

acf63a56d5f50cfaa23f57ee52a66df261ec241ce13fd81ccac39a3bbbddd3ff;

the latter's final typeset source has SHA256

7a7733e0d6910398b6de716604a82ef5287a55002e1874327490a70ccc3dd414.

Both complete proofs are retained beside this continuation. The analytic theta section and critical quotient retain the full Hochschild comparison (H1)–(H42) used there. All tensor completions below are the same completed projective tensor products as in (PC32).

### 4.1 The original source, two actual targets, and their inverses

Retain the original nonempty packet  $h$ ,  $k \geq 2$ , and  $L > 0$  for the positive-weight all-integer circle realization. The final paragraph of this subsection gives the exact  $k = 1$  support modification. Write  $Z$  for the original selected zero set, with its complete multiplicities  $m_\rho$ . Put

$$\begin{aligned} R &= \mathbb{C}[S], & E &= R/(\chi), & q &= \deg \chi, & A &= M_S, & e_0 &= [1]_\chi, \\ \chi(S) &= \prod_{\lambda} (S - \lambda)^{m_\lambda}, & m_\lambda &= \max_{\substack{\rho_1 + \dots + \rho_k = \lambda \\ \rho_i \in Z}} \left( 1 + \sum_{i=1}^k (m_{\rho_i} - 1) \right). \end{aligned} \quad (\text{DC1})$$

Every tuple is ordered. Equal sums are combined only by their literal equality in  $\mathbb{C}$ . In the tuple-local algebra  $\mathbb{C}[x_1, \dots, x_k]/(x_i^{m_{\rho_i}})$ , the highest nonzero power of  $x_1 + \dots + x_k$  has degree  $D = \sum_i (m_{\rho_i} - 1)$  and coefficient  $D! / \prod_i (m_{\rho_i} - 1)!$  on  $\prod_i x_i^{m_{\rho_i} - 1}$ . All higher powers vanish. Taking the maximum over equal sums proves (DC1), including every exponent. The nonempty original packet gives  $q \geq 1$ .

Use only tuples in  $Z_c = \{\rho \in Z : \text{Re } \rho = 1/2\}$  to define  $a = \chi_{c,k}$ . At an ambient root  $\lambda$  let  $r_\lambda$  be the same maximum over those all-critical tuples, and set  $r_\lambda = 0$  if there is none. Thus

$$a(S) = \prod_{\lambda} (S - \lambda)^{r_\lambda}, \quad 0 \leq r_\lambda \leq m_\lambda, \quad a \mid \chi. \quad (\text{DC2})$$

If  $Z_c$  is empty,  $a = 1$ . This definition retains the full nilpotent length of the all-critical cyclic image.

The actual tensor-critical map is

$$C = \Gamma^{\widehat{\otimes} k} \sigma_h^{\widehat{\otimes} k} \eta_k : E \longrightarrow Q_{\text{crit}}^{\widehat{\otimes} k}, \quad \eta_k = M_{v_h^{\otimes k}} \alpha_k, \quad v_h = j_h((2\xi)/h). \quad (\text{DC3})$$

Its full map type includes  $\iota_1 : \mathbb{C}[S]/h(S) \rightarrow E_h = \mathbb{C}[s]/h(s)$ ,  $[P(S)] \mapsto [P(s)]$ , and  $\bar{\alpha}_k = (\iota_1^{-1})^{\otimes k} \alpha_k$ . Consequently  $\eta_k = \eta_h^{\otimes k} \bar{\alpha}_k$  with  $\eta_h = M_{v_h} \iota_1$ . Write  $q_{\mathcal{B}} : \mathcal{B} \rightarrow Q = \mathcal{B}/\Theta V$  for the original quotient map. The original source comparison is still  $q_{\mathcal{B}}^{\widehat{\otimes} k} \mathcal{V}_{h,k} = \sigma_h^{\widehat{\otimes} k} \eta_k \pi_\chi$ . No arithmetic unit is replaced by 1.

By (PC31)–(PC35),  $\ker C = (a)/(\chi)$ . Set  $W_a = \text{im } C$ . The induced map and its actual inverse are

$$\begin{aligned} j_a : R/(a) &\xrightarrow{\sim} W_a, & j_a[P]_a &= C[P]_\chi, \\ j_a^{-1} &= \alpha_{c,k}^{-1} \mathcal{R}_c^{\widehat{\otimes} k} \Big|_{W_a}, & C &= j_a \pi_a. \end{aligned} \quad (\text{DC4})$$

Here  $\pi_a : E \rightarrow R/(a)$  is polynomial reduction, and  $\alpha_{c,k} : R/(a) \hookrightarrow E_c^{\otimes k}$  is the injection  $[P] \mapsto P(A_c^{[k]})e_c^{\otimes k}$ , with  $e_c$  the identity of the original critical CRT ideal. The inverse in (DC4) has domain the image of this injection, because (PC33) proves  $\mathcal{R}_c^{\widehat{\otimes} k} C = \alpha_{c,k} \pi_a$ . In each one-factor critical coordinate,

$$\Xi_\rho[f] = \sum_{j=0}^{m_\rho-1} \frac{i^{-j}}{j!} f^{(j)}(\text{Im } \rho) x^j, \quad \mathcal{R}_c = M_{\varepsilon_c} \Xi_c, \quad \varepsilon_c v_c = e_c. \quad (\text{DC5})$$

Thus (DC4) retains the full tensor product of the factors  $i^{-j}/j!$  and the recursively inverted original local units. It is an isomorphism of finite-dimensional  $R$ -modules. The action on  $W_a$  is the restriction  $M_a$  of the original sum multiplier  $M^{[k]}$ , and  $M_a j_a = j_a M_S$ . No algebra multiplication on the ambient critical quotient is needed.

For the circle retain the original frequencies, scalar measure, and vector amplitude:

$$\begin{aligned} S_n &= k/2 + 2\pi i n/L, & \nu_n &= (2\pi/L) m_{h,k}(2\pi n/L) > 0, \\ b &= p_L = \prod_{n \in \mathcal{Z}_L} (S - S_n), & \mathcal{Z}_L &= \{n \in \mathbb{Z} : \chi(S_n) = 0\}, \\ \beta_n &= \widehat{\mathcal{U}_k F_h^{\otimes k}}(2\pi n/L), & \|\beta_n\|_K^2 &= 2\pi m_{h,k}(2\pi n/L). \end{aligned} \quad (\text{DC6})$$

The frequencies are distinct, so  $b$  is squarefree and divides  $\chi$ ; an empty product is 1. In the actual circle Fourier basis  $L^{-1/2} e^{-2\pi i n r/L}$ , the polynomial source has coefficient  $L^{-1/2} \beta_n P(S_n)$ . The scalar-to-vector map  $c \mapsto (L^{-1/2} \beta_n c_n)_n$  has squared norm  $\sum_n \nu_n |c_n|^2$ , proving the isometry with the weighted cyclic space without rescaling any  $\beta_n$  or  $\nu_n$ .

For completeness, the Hilbert quotient in (P56) is exactly

$$W_b = \ell^2(\mathbb{Z}, \nu) / \overline{\{(\chi(S_n) P(S_n))_n : P \in R\}} \xrightarrow{\sim} \bigoplus_{n \in \mathcal{Z}_L} \mathbb{C}, \quad \|v\|_{W_b}^2 = \sum_{n \in \mathcal{Z}_L} \nu_n |v_n|^2. \quad (\text{DC7})$$

Indeed the relation closure vanishes on  $\mathcal{Z}_L$ . For the reverse inclusion divide a vector vanishing there by  $\chi(S_n)$  off that finite set; the result belongs to  $\ell^2(|\chi|^2 \nu)$ . Polynomial density in this space follows from the original exponential moment (P54): an orthogonal vector defines a finite complex measure with analytic Fourier transform in a strip, whose derivatives at zero all vanish. The power series, the identity theorem, and Fourier uniqueness force the vector to vanish. Multiplying the approximating polynomials by  $\chi$  proves the reverse inclusion. This also proves the stated isometric quotient. All finite vectors in (DC7) are attained by polynomial interpolation. The actual observation, its injection, and inverse are therefore

$$\begin{aligned} \theta &= \theta_L : E \rightarrow W_b, & \theta[P]_\chi &= (P(S_n))_{n \in \mathcal{Z}_L}, & \ker \theta &= (b)/(\chi), \\ j_b &: R/(b) \xrightarrow{\sim} W_b, & j_b[P]_b &= (P(S_n))_{n \in \mathcal{Z}_L}, & \theta &= j_b \pi_b, \\ j_b^{-1}(v) &= \left[ \sum_{n \in \mathcal{Z}_L} v_n \prod_{\substack{m \in \mathcal{Z}_L \\ m \neq n}} \frac{S - S_m}{S_n - S_m} \right]_b. \end{aligned} \quad (\text{DC8})$$

Every denominator is nonzero. The action  $M_b$  on  $W_b$  is  $(v_n) \mapsto (S_n v_n)$ , so  $M_b j_b = j_b M_S$  and  $\theta A = M_b \theta$ . The class  $j_b[1]_b$  has squared norm  $\sum_{n \in \mathcal{Z}_L} \nu_n$ , which is positive exactly when  $\mathcal{Z}_L$  is nonempty and is zero otherwise. This actual number has not been assigned the value one. For  $k=1$  the identical proof uses only the actual support  $I_L = \{n : w_h(2\pi n/L) > 0\}$  and  $\mathcal{Z}_L = \{n \in I_L : \chi(S_n) = 0\}$  in (DC6)–(DC8). All formulas below use that support-specific  $b$  and remain exact. For the actual full-order packet, this restriction omits no sampled root of  $h$ . At such a root  $\rho$  write  $g(s) = (s - \rho)^{m_\rho} g_\rho(s)$

and  $h(s) = (s - \rho)^{m_\rho} h_\rho(s)$ , where  $g_\rho(\rho)h_\rho(\rho) \neq 0$  by the original complete-order hypothesis. Hence  $v_h(\rho) = g_\rho(\rho)/h_\rho(\rho) \neq 0$ . For  $\rho = 1/2 + 2\pi in/L$  the actual weight is  $\nu_n = |v_h(\rho)|^2/L > 0$ , by  $w_h(t) = |v_h(1/2 + it)|^2/(2\pi)$ . Thus for  $k = 1$  the polynomial  $b$  is also exactly the product over all sampled roots of  $\chi = h$ , even if some other lattice atoms have zero weight. No positivity of those other atoms is used.

## 4.2 The common quotient and the actual joint image

For any monic divisor  $s \mid \chi$ , denote the reduction by  $\pi_s : E \rightarrow E_s = R/(s)$  and its kernel by  $K_s = (s)/(\chi)$ . In particular  $E_1 = 0$ ,  $K_1 = E$ ,  $E_\chi = E$ , and  $K_\chi = 0$ . Put

$$d = \gcd(a, b), \quad l = \text{lcm}(a, b) = ab/d, \quad d \mid a, b \mid l \mid \chi. \quad (\text{DC9})$$

All gcds and lcms are monic. Define the maps into the common quotient  $E_d$  by their full types:

$$f_a : W_a \rightarrow E_d, \quad f_a = \pi_d^a j_a^{-1}, \quad f_b : W_b \rightarrow E_d, \quad f_b = \pi_d^b j_b^{-1}, \quad f_a C = f_b \theta = \pi_d. \quad (\text{DC10})$$

Here  $\pi_d^a : E_a \rightarrow E_d$  and  $\pi_d^b : E_b \rightarrow E_d$  are reduction. Both maps are onto, commute with  $S$ , and are continuous on these finite-dimensional spaces. They give the actual fibre product

$$W_\vee = \{(x, y) \in W_a \oplus W_b : f_a x = f_b y\}, \quad J = (C, \theta) : E \rightarrow W_\vee. \quad (\text{DC11})$$

Thus the compatibility condition is expressed through the exact weighted inverse (DC4) and original-circle inverse (DC8).

There is an explicit isomorphism

$$\begin{aligned} j_l : E_l &\xrightarrow{\sim} W_\vee, & j_l[P]_l &= (j_a[P]_a, j_b[P]_b), & J &= j_l \pi_l, \\ \ker J &= K_l, & \dim W_\vee &= \deg l = \deg a + \deg b - \deg d. \end{aligned} \quad (\text{DC12})$$

Well-definedness follows from  $a, b \mid l$ . If both components vanish,  $a$  and  $b$  divide  $P$ ; their prime-power factorizations show that  $l$  divides  $P$ . This proves injectivity. To prove surjectivity with an explicit inverse, write  $a = da'$ ,  $b = db'$  and compute polynomials  $u, v$  with  $ua' + vb' = 1$ . Given compatible representatives  $X$  modulo  $a$  and  $Y$  modulo  $b$ ,  $Y - X = dT$  for some  $T \in R$ . Then

$$P = X + auT, \quad P - X = auT, \quad P - Y = -bvT. \quad (\text{DC13})$$

It has the required two residues. A different choice of representatives, Bezout coefficients, or  $T$  changes  $P$  by a multiple of both  $a$  and  $b$ , hence of  $l$ . This proves the well-defined inverse and the entire isomorphism. The Euclidean algorithm terminates because its successive nonzero remainders have strictly decreasing degrees, and its back substitution gives  $u, v$ . The same proof covers  $a = 1$ ,  $b = 1$ , or  $d = 1$ .

The complete exact quotient sequence is

$$0 \longrightarrow E_l \xrightarrow{[P] \mapsto ([P]_a, [P]_b)} E_a \oplus E_b \xrightarrow{(x, y) \mapsto \pi_d^a x - \pi_d^b y} E_d \longrightarrow 0. \quad (\text{DC14})$$

The difference map is onto because its first component is onto; its kernel and first-map injectivity were just proved. Transport by  $j_a, j_b, j_l$  proves the identical sequence for the actual images. In parallel, the kernel diamond has

$$\begin{aligned} K_a \cap K_b &= K_l, & K_a + K_b &= K_d, \\ 0 \longrightarrow K_l &\xrightarrow{z \mapsto (z, -z)} K_a \oplus K_b \xrightarrow{(x, y) \mapsto x + y} K_d \longrightarrow 0. \end{aligned} \quad (\text{DC15})$$

The intersection follows from the same divisibility argument. Bezout applied to  $a, b$  gives  $d = ua + vb$ , which proves the equality of the sum with  $K_d$ . The displayed signs then prove exactness. Every map in (DC14)–(DC15) commutes with multiplication by  $S$ .

In particular  $E_d = E/(K_a + K_b)$  is the largest common quotient of the original two observations in the following exact sense. For any vector space  $T$  and linear maps  $g_a : W_a \rightarrow T$ ,  $g_b : W_b \rightarrow T$  with  $g_a C = g_b \theta$ , their common composite vanishes on  $K_a + K_b$ . It therefore has a unique factor  $g_d : E_d \rightarrow T$ , defined by  $g_d[P]_d = g_a C[P]_\chi$ . It is well defined by that vanishing and satisfies  $g_a = g_d f_a$ ,  $g_b = g_d f_b$  by surjectivity. If the given maps intertwine specified actions, so does  $g_d$ , by applying the surjective map  $\pi_d$ . This is the common-quotient assertion with its domain and universality fully specified.

### 4.3 Every local length and both exact factorization criteria

At a root  $\lambda$  of  $\chi$ , use  $x = S - \lambda$  and retain

$$\begin{aligned} \chi &= x^{m_\lambda} c_\lambda(x), \quad a = x^{r_\lambda} a_\lambda(x), \quad b = x^{s_\lambda} b_\lambda(x), \\ c_\lambda(0) a_\lambda(0) b_\lambda(0) &\neq 0, \quad s_\lambda \in \{0, 1\}, \\ d &= x^{\min(r_\lambda, s_\lambda)} d_\lambda(x), \quad l = x^{\max(r_\lambda, s_\lambda)} l_\lambda(x), \quad d_\lambda(0) l_\lambda(0) \neq 0. \end{aligned} \tag{DC16}$$

These are the literal polynomial cofactors at  $\lambda$ , with no chosen constant rescaling. The source local ring is  $R_\lambda = \mathbb{C}[x]/(x^{m_\lambda})$ . For  $s \in \{a, b, d, l\}$ , write its exponent as  $e_\lambda(s)$  and its cofactor as  $s_\lambda^{\text{unit}}(x)$ . The exact kernel inclusion is

$$\mathbb{C}[x]/(x^{m_\lambda - e_\lambda(s)}) \longrightarrow R_\lambda, \quad [B] \longmapsto [x^{e_\lambda(s)} s_\lambda^{\text{unit}}(x) B]. \tag{DC17}$$

It is injective and its image is precisely the local component of  $K_s$ . Indeed the unit is invertible by recursive coefficient division; multiplying by that inverse and cancelling the stated power of  $x$  proves injectivity. Its image is the principal ideal by definition. The quotient is the reduction  $R_\lambda \rightarrow \mathbb{C}[x]/(x^{e_\lambda(s)})$ . All these maps intertwine the actual action  $\lambda + x$ . Thus the four quotient lengths are respectively  $r_\lambda, s_\lambda, \min(r_\lambda, s_\lambda)$ , and  $\max(r_\lambda, s_\lambda)$ , and the four kernel lengths are  $m_\lambda$  minus those lengths. No primary component or nilpotent jet is discarded from the source.

There is a map of the actual images  $T_{ab} : W_a \rightarrow W_b$  with  $\theta = T_{ab} C$  if and only if  $b \mid a$ . In that case it is unique and its complete formula is

$$T_{ab} = j_b \pi_b^a j_a^{-1}, \quad M_b T_{ab} = T_{ab} M_a. \tag{DC18}$$

Necessity follows by applying the assumed factorization to  $K_a$ : it gives  $K_a \subset K_b$ . Since the class of  $a$  is in  $K_a$ , this means  $a = bP + \chi Q$  for some  $P, Q$ , and  $b \mid \chi$  then gives  $b \mid a$ . Sufficiency is the displayed reduction map; uniqueness follows because  $C$  is onto  $W_a$ . Intertwining follows from the corresponding property of every factor. The reverse factorization exists if and only if  $a \mid b$ , and is exactly

$$T_{ba} = j_a \pi_a^b j_b^{-1}, \quad C = T_{ba} \theta. \tag{DC19}$$

The same proof establishes it even if an image is zero. Consequently the two observations have mutually inverse factorizations over  $E$  if and only if  $a = b$ . Since  $b$  is squarefree, (DC19) requires every critical cyclic exponent to be at most one and every such root to be actually sampled. Formula (DC18) instead requires each sampled root to occur in  $a$ ; it places no upper bound on its critical multiplicity.

Their extra observations are calculated by the original maps:

$$\begin{aligned} \theta(K_a) &\simeq K_a/K_l \simeq [a]_b E_b, & \dim \theta(K_a) &= \deg b - \deg d, \\ C(K_b) &\simeq K_b/K_l \simeq [b]_a E_a, & \dim C(K_b) &= \deg a - \deg d. \end{aligned} \tag{DC20}$$

Here  $[a]_b E_b$  is the ideal generated by the residue of  $a$  in  $R/(b)$ , and  $[b]_a E_a$  is the ideal generated by the residue of  $b$  in  $R/(a)$ . Explicitly the first isomorphism sends the class of  $aP$  modulo  $K_l$  to  $j_b[aP]_b$ . Its kernel is exactly  $K_a \cap K_b$ , and all values in that ideal are attained. The ideal in  $R/(b)$  is also the ideal generated by  $d$ , by Bezout and reduction modulo  $b$ , so its dimension is  $\deg b - \deg d$ . Interchanging  $a, b$  proves the second line. Alternatively (DC17) calculates the local lengths as  $s_\lambda - \min(r_\lambda, s_\lambda)$  and  $r_\lambda - \min(r_\lambda, s_\lambda)$ .

This gives a concrete criterion for a circle value of a class killed by  $C$ . It occurs exactly at a sampled  $\lambda$  with  $s_\lambda = 1$  and  $r_\lambda = 0$ . Let  $e_\lambda \in E$  be the CRT idempotent equal to 1 in the  $\lambda$ -primary ring and 0 in every other primary ring. Polynomial Bezout produces this idempotent in the original  $E$ . Then

$$C e_\lambda = 0, \quad \theta e_\lambda = (\delta_{\lambda, S_n})_{n \in \mathcal{Z}_L}, \quad \|\theta e_\lambda\|^2 = \nu_n \quad (S_n = \lambda). \quad (\text{DC21})$$

Such a  $\lambda$  has real part  $k/2$  because it is sampled. Its absence from  $a$  means precisely that no all-critical tuple has this sum; the ambient tuples realizing it have at least one off-critical member. The statement is a criterion and explicit map for actual roots of the retained packet; it asserts no existence of an off-critical zero. When  $r_\lambda \geq 1$  and  $s_\lambda = 1$ , the circle sees the constant critical jet and all  $r_\lambda - 1$  higher critical jets lie in  $C(K_b)$ . If mixed tuples raise  $m_\lambda$  above  $r_\lambda$ , the additional  $m_\lambda - r_\lambda$  source jets remain in  $K_a$  and  $K_l$ . For an unsampled root every surviving critical jet lies in  $C(K_b)$ . These are all local possibilities, directly from (DC16)–(DC20).

#### 4.4 The same period transport and all four action defects

Keep the period matrix  $\Pi(t) = \Pi_k(t)$  of the original polynomial  $\chi$  from (PC37), with its original ordered rays,  $u \in \mathbb{C}^*$ , and increasing-power basis. In particular its entries are the integrals of  $S^j dS$  against  $\exp((S^{q+1}/(q+1) + \sum_{\ell=0}^{q-1} c_\ell S^{\ell+1}/(\ell+1) - tS)/u)$  on those rays. It is invertible for every  $t$ , including repeated-root parameters, by the complete fixed-ray and determinant proof (PC16)–(PC18). Retain

$$\ell[P]_\chi = [S^{q-1}] \text{rem}_\chi P, \quad R_0 = e_0 \ell, \quad A_t = A + t R_0, \quad \Pi' = -\Pi A_t / u, \quad A_t^{\text{per}} = \Pi A_t \Pi^{-1}. \quad (\text{DC22})$$

Here  $R_0$  is an endomorphism and is distinct from the polynomial ring  $R$  used above. The following indexed families use four formal roles  $\mathcal{I} = \{\text{critical}, \text{circle}, \text{common}, \text{joint}\}$  with associated polynomials  $(s_i)_{i \in \mathcal{I}} = (a, b, d, l)$ . Subscripts  $a, b, d, l$  on the actual targets and observations below are shorthand for these four roles. In  $K_s$ ,  $\deg s$ , and polynomial reduction,  $s$  means the associated polynomial. Thus even if two or all four polynomials coincide, their four actual target realizations and maps remain separately specified. Set  $W_d = E_d$ ,  $j_d = I$ ,  $T_d = \pi_d$ ,  $T_a = C$ ,  $T_b = \theta$ ,  $T_l = J$ , and  $W_l = W_\vee$ . Let  $M_s$  be multiplication by  $S$  on  $W_s$  through  $j_s$ . For  $s = l$  it is explicitly  $(M_a, M_b)|_{W_\vee}$ ; this restriction is well defined by (DC10). For all four indices,  $T_s = j_s \pi_s$ ,  $T_s A = M_s T_s$ , and  $\ker T_s = K_s$ .

Define the four actual period observations on the same target coordinate space  $\mathbb{C}^q$  by

$$T_{s,t} = T_s \Pi^{-1}, \quad \mathcal{K}_{s,t} = \ker T_{s,t} = \Pi K_s, \quad s \in \{a, b, d, l\}. \quad (\text{DC23})$$

They are onto their fixed images, with holomorphic kernels and constant ranks  $\deg s$ . If  $j_s^{-1}$  is followed by any polynomial representative of degree less than  $\deg s$ , its composition with  $\Pi$  is an explicit holomorphic right inverse of  $T_{s,t}$ . The diagram remains exactly the same:

$$f_a T_{a,t} = f_b T_{b,t} = T_{d,t}, \quad T_{l,t} = (T_{a,t}, T_{b,t}), \quad \mathcal{K}_{a,t} \cap \mathcal{K}_{b,t} = \mathcal{K}_{l,t}, \quad \mathcal{K}_{a,t} + \mathcal{K}_{b,t} = \mathcal{K}_{d,t}. \quad (\text{DC24})$$

Every equality follows by composing its previously proved source equality with the invertible map  $\Pi^{-1}$ . For the transported original action  $B_t = \Pi A \Pi^{-1}$  this is an exact diagram of  $R$ -modules:

$T_{s,t}B_t = M_sT_{s,t}$ . Indeed multiplication cancels the two middle factors  $\Pi^{-1}\Pi$  and gives the original identity  $T_sA = M_sT_s$ . The deformed action  $A_t^{\text{per}}$  instead has the following fully calculated defects.

Before restricting any domain, the complete action defect is

$$\boxed{T_{s,t}A_t^{\text{per}} - M_sT_{s,t} = tT_s(e_0)\ell\Pi^{-1}}. \quad (\text{DC25})$$

Indeed the first term is  $T_s(A + te_0\ell)\Pi^{-1}$ ; the  $A$  part is  $M_sT_s\Pi^{-1}$  by the original intertwining. This proves the formula without asserting invariance of a deformed kernel. The weights in its four target vectors are explicit:

$$T_a(e_0) = C(e_0), \quad T_b(e_0) = (1)_{n \in \mathcal{Z}_L}, \quad T_d(e_0) = [1]_d, \quad T_l(e_0) = (C(e_0), (1)_{n \in \mathcal{Z}_L}). \quad (\text{DC26})$$

The first vector has the raw jet recovery  $(M_{v_c})^{\otimes k} \alpha_{c,k}[1]_a$  before applying the inverse units; the second has its original norm from (DC8). If  $\deg s > 0$ ,  $T_s(e_0) \neq 0$  because  $j_s$  is injective. If  $\deg s = 0$ , its target and every displayed defect are zero.

For any monic  $v \mid \chi$  with  $\deg v < q$ , the ideal  $K_v$  has the exact basis  $[v], [Sv], \dots, [S^{q-\deg v-1}v]$ . Monicity gives different leading degrees, proving independence; division by  $\chi/v$  proves spanning. Its last member has degree  $q-1$  and leading coefficient 1, so  $\ell|_{K_v}$  is onto. Consequently, for  $t \neq 0$ ,  $\deg s > 0$ , and  $\deg v < q$ , restriction of the  $s$ -defect to  $\mathcal{K}_{v,t}$  has

$$\begin{aligned} \text{im } (T_{s,t}A_t^{\text{per}} - M_sT_{s,t})|_{\mathcal{K}_{v,t}} &= \mathbb{C}T_s(e_0), \\ \ker (T_{s,t}A_t^{\text{per}} - M_sT_{s,t})|_{\mathcal{K}_{v,t}} &= \Pi(K_v \cap \ker \ell), \quad \text{rank} = 1. \end{aligned} \quad (\text{DC27})$$

For  $v = \chi$  the restricted domain is zero; for  $t = 0$  or  $s = 1$  the whole restricted defect is zero. In particular the self-kernel defects have rank one precisely when  $t \neq 0$  and  $0 < \deg s < q$ . Formula (DC27) also calculates the critical defect on the circle kernel and the circle defect on the critical kernel, with their full domains. The observations themselves on those crossed kernels retain (DC20):  $T_{b,t}(\mathcal{K}_{a,t}) = \theta(K_a)$  and  $T_{a,t}(\mathcal{K}_{b,t}) = C(K_b)$ .

The factorizations (DC18)–(DC19), whenever their proved divisibility criterion holds, remain unchanged after period transport. Conversely a factorization at even one parameter  $t$  can be composed with  $\Pi(t)$  and gives the original factorization, so its same divisibility criterion is necessary. The maps  $f_a, f_b$  carry the first two defects to the common defect, and the joint defect has the two original defects as components, because  $f_aC(e_0) = f_b\theta(e_0) = [1]_d$ . Thus both the observations and their entire failure of deformed-action intertwining commute with the exact diamond.

## 4.5 The exact extra circle summand

Squarefreeness of the actual circle polynomial gives a more explicit joint-image formula. Put  $b_{\text{new}} = b/d$ . Its roots are exactly the sampled roots absent from  $a$ , so  $\gcd(a, b_{\text{new}}) = 1$  and  $l = ab_{\text{new}}$ . Denote by  $W_{\text{new}}$  the actual circle coordinate subspace on these roots, with its inherited metric  $\sum_{S_n: b_{\text{new}}(S_n)=0} \nu_n |v_n|^2$  and diagonal action  $S_n$ . There is an exact isomorphism

$$\begin{aligned} W_{\vee} &\xrightarrow{\sim} W_a \oplus W_{\text{new}}, & (x, y) &\longmapsto (x, y_{\text{new}}), \\ (x, z) &\longmapsto (x, y), & y_n &= \begin{cases} (j_a^{-1}x)(S_n), & d(S_n) = 0, \\ z_n, & b_{\text{new}}(S_n) = 0. \end{cases} \end{aligned} \quad (\text{DC28})$$

Evaluation in the first case is independent of a polynomial representative because  $a(S_n) = 0$ . It gives exactly the shared reduction in (DC10); the two cases exhaust the sampled roots. Thus the displayed maps are mutually inverse. They intertwine the original actions, since evaluation of  $SP$  gives  $S_nP(S_n)$ .

This is the actual-image transport of the CRT isomorphism  $R/(l) \simeq R/(a) \oplus R/(b_{\text{new}})$ . Its first factor retains the original unit-weighted critical map and all of its nilpotent jets; its second factor retains the actual new circle amplitudes and weights. It asserts an isomorphism of finite-dimensional  $R$ -modules. If one assigns the actual circle norm to the second component of  $W_{\vee} \subset W_a \oplus W_b$ , that norm under (DC28) is exactly  $\sum_{d(S_n)=0} \nu_n |(j_a^{-1}x)(S_n)|^2 + \sum_{b_{\text{new}}(S_n)=0} \nu_n |z_n|^2$ . The shared first sum remains present, so the module isomorphism has not silently been asserted to preserve a chosen direct-sum metric on  $W_a \oplus W_{\text{new}}$ .

The new circle coordinates have an explicit original polynomial source lift. For a prescribed vector  $z = (z_n)_{b_{\text{new}}(S_n)=0}$ , set

$$P_z(S) = a(S) \sum_{b_{\text{new}}(S_n)=0} \frac{z_n}{a(S_n)} \prod_{\substack{b_{\text{new}}(S_m)=0 \\ m \neq n}} \frac{S - S_m}{S_n - S_m}. \quad (\text{DC29})$$

Each  $a(S_n)$  and each  $S_n - S_m$  in a denominator is nonzero, by the proved coprimality and distinct sampled frequencies. The factor  $a$  puts  $[P_z]_{\chi}$  in  $K_a$ . At a new sample the interpolation product gives exactly  $z_n$ ; at a shared sample the factor  $a$  gives zero. Consequently  $C[P_z]_{\chi} = 0$  and  $\theta[P_z]_{\chi} = (0, z)$ , with squared circle norm  $\sum \nu_n |z_n|^2$  on precisely the new samples. At source level the lift is the unchanged  $\mathcal{V}_{h,k} P_z$ , whose cohomology class is  $\sigma_h^{\otimes k} \eta_k [P_z]_{\chi}$ . Thus the circle summand in (DC28) has an actual unit-weighted arithmetic source, including every coefficient  $a(S_n)^{-1}$ . When there are no new samples, the sum and its lift are zero. For  $k = 1$ , every sampled root of  $\chi = h$  is itself critical, so  $b \mid a$  and this extra circle summand is always zero.

The differential equation of all four transported observations retains the same forcing as (DC25):

$$\boxed{u \partial_t T_{s,t} - M_s T_{s,t} = t T_s(e_0) \ell \Pi^{-1}, \quad s \in \{a, b, d, l\}.} \quad (\text{DC30})$$

Differentiating  $\Pi^{-1} \Pi = I$  and using (DC22) gives  $(\Pi^{-1})' = A_t \Pi^{-1} / u$ . Multiplication by  $u T_s$  and the original relation  $T_s A = M_s T_s$  prove (DC30). The fixed maps  $f_a, f_b$  carry these differential equations to the common one by (DC10); the joint equation is their paired equation. The original factors  $u, t$ , and the full four vectors in (DC26) remain in both formulas.

All coefficient arrows above have their original supported Split-Zero lifts:  $v^{\bullet} \mapsto (Tv)^{\bullet}$  and  $\tau \mapsto \tau$ , with the specified support retained. The kernel inclusions (DC15), local maps (DC17), and period kernels (DC23) identify exactly the represented classes sent to the receiving supported zero. Projection to ordinary coefficients commutes with every displayed arrow. The common quotient and joint image therefore preserve the same support data while recording these different, explicitly calculated coefficient kernels.